MILITARY SPECIFICATION

STARTER/AUXILIARY POWER UNIT,
AIRCRAFT ENGINE, GAS TURBINE
TYPE, DETAILED SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and Agencies of the Department of Defense.

1. SCOPE

This specification covers a gas turbine starter/APU for starting the main aircraft engine and providing emergency electrical power on the AV-8B aircraft. This specification also establishes the content and format to be used by the supplier for the preparation of the gas turbine starter/APU model specification.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents, of the issues in effect on the date of invitation for bids or request for proposals, are a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirewound (For Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated-Plywood

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2995

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PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-T-60	Packaging, Waterproof

Military

MIL-P-116	Preservation, Packaging, Method of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation Grades JP-4 and JP-5
MIL-C-6021	Casting, Classification and Inspection of
MIL-H-6088	Heat Treatment of Aluminum Alloys
MIL-I-6866	Inspection, Pentrant Method of
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol AMPT, General Requirements for
MIL-F-7179	Finishes and Coatings; Protection of Aerospace Weapons Systems, Structures and Parts
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-L-7808G	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for

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MIL-P-9400	Plastic Laminate and Sandwich Construction, Aircraft Structural Process Parts Specification Requirements
MIL-C-11796	Corrosion Preventative Compound, Petrolatum, Hot Application
MIL-C-16173	Corrosion Preventative Compound, Solvent Cutback, Cold Application
MIL-A-21180	Aluminum Alloy Castings, High Strength
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base
MIL-E-25499	Electrical System, Aircraft, Design and Installation of, General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environment Resistant, Removable Crimp Contacts

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-453	Inspection, Radiographic
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipment)
DOD-STD-480	Configuration Control, Engineering Changes, Deviations and Waivers

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MIL-STD-704	Aircraft Electric Power, Characteristics
MIL-STD-721	Definition of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Method
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1523	Age Control of Age Sensitive Elastomeric Material
MIL-STD-2076	Unit Under Test Compatibility With Automatic Test Equipment; General Requirements For
MIL-STD-2077	Test Program Sets; General Requirements For
MS33666	Packing, Preformed - Aeronautical, Elastomeric, Range of Sizes
MS33668	Packing, Preformed - Tube, Fitting Elastomeric, Range of Sizes

PUBLICATIONS (Air Force-Navy Aeronautical Bulletins)

MIL-BULL-343	Documents Applicable to Aircraft Engines and Propellers - Use of
MIL-BULL-544	List of Specifications and Standards (Book Form) Approved by the Naval Air Systems Command

PUBLICATIONS (NAVAIR Requirements)

AR- 10	Maintainability of Avionics Requirements and Systems, General Requirements for

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Request for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

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2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1-1962	Surface Texture

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

SOCIETY OF BRITISH AEROSPACE COMPANIES, LTD.

RS263	Accessory Drives and Mounting Pads (Taper Flange and Involute Splines)

(Application for copies should be addressed to the Society of British Aerospace Companies, LTD., Technical Section, 29 King Street, London SW1Y 6RD, England, U.K.)

3. REQUIREMENTS

3.1 Precedence. The contract shall have precedence over any specification. Any deviation from this specification or from subsidiary specifications or standards, where applicable, is subject to approval in writing by the procuring activity.

3.2 First Article. The parts furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.2). The Naval Air Propulsion Center (PE61), Trenton, New Jersey 08628, is the activity responsible for approving equipment submitted under this specification.

3.3 Dimensions. The aircraft engine gas turbine starter/APU components defined in 3.6.3 shall conform-to the dimensions shown in Figures 1A through 9. Where only maximum or minimum dimensions are shown, the starter need not have the shape shown, but the starter, including all protrusions, shall be contained within the outline shown. Design details which are shown but are not specifically required are shown as examples and for information only; such details are optional.

3.4 Data. Unless otherwise specified in the contract, data cited in this specification is only required as part of the first article submittal.

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3.4.1 Model Specification. A model specification in accordance with 6.3 shall be prepared by the supplier and submitted to NAVAIR (AIR-536) for approval.

3.5 Materials and Parts. The limitations and restrictions placed on the use of materials and components by MIL-BULL-544 shall pertain. Materials, parts, and processes shall conform to applicable Government specifications listed in MIL-BULL-544. Contractor or industry specifications may be used for materials or processes not listed in MIL-BULL-544, provided the specifications are approved by the approving activity and contain provisions for adequate tests.

3.5.1 Materials.

3.5.1.1 Toxic Products. Materials which are toxic in themselves, or which become toxic when pulverized or vaporized, or which have toxic products of combustion, shall be used only upon written permission of the approving activity.

3.5.1.2 Non-Metals. Non-metallic materials used, including plastics, fabrics, and protective finishes shall be flame-resistant, shall not support combustion, shall be moisture resistant, and shall not be adversely affected by weather, temperature, and ambient conditions.

Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used, all nutrient materials and associated fungicidal treatments used shall be specified on the starter/APU manufacturing drawings.

Elastomeric components shall be fabricated from materials covered by applicable specifications and shall have maximum practicable ozone and aging resistance and hydrolytic stability consistent with performance requirements. All elastomeric materials used shall be subject to the specific acceptance by approving activity based upon the demonstrated suitability of the materials for the expected operating and storage environments. Age controls for synthetic rubber parts shall be in accordance with MIL-STD-1523.

Process specifications for adhesive-bonded construction including reinforced plastic, sandwich, and foamed-in-place core construction shall be prepared in accordance with MIL-P-9400. All sandwich construction shall be sealed or otherwise designed to preclude entrance or entrapment of water and other contaminants. Perforated metallic core shall not be used. When metallic core is used, it shall be corrosion resistant.

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All reinforced plastic construction shall be in conformance with the contractor's process specifications, prepared in accordance with MIL-P-9400 and accepted by the approving activity.

3.5.1.3 Metals. All metals shall be compatible with the intended temperature, functional, service and storage conditions to which the starter/APU will be exposed, Metals which are not protected from corrosion by lubricating oil shall be of the corrosion-resisting type, or shall be suitably treated to resist corrosion in accordance with 3.5.1.4.

3.5.1.3.1 Steels (except corrosion-resistant).

Aircraft quality, vacuum-melted steel shall be used for parts which are heat treated to an ultimate tensile strength 220,000 psi and above. Such applications shall be subject to specific acceptance by the approving activity.

b. The ultimate tensile strength range in production parts shall not exceed the following limits:

Minimum Ultimate Tensile Strength P.S.I.	Permissible Variation, P.S.I.
220,000	-0 + 20,000
260,000	-0 + 20,000
270,000	-0 + 20,000
280,000	-0 + 20,000

c. Preference shall be given, in selection of carbon and low alloy steels, to compositions having the least hardenability which will insure thorough-hardening of the part concerned.

d. Compositions shall be selected such that heat treatment to the required strength and service temperatures shall preclude temper embrittlement.

Steels shall be selected having ductile-brittle fracture transition temperatures as determined by impact test below the minimum operating temperature.

f. Steels whose mechanical properties are developed by cold deformation shall be so selected that the recovery temperature will be at least 50°F (28°C) above the expected operating temperature range.

g. Critical parts shall be designed and processed so as to result in no decarburization of highly stressed areas. Designs

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shall preclude use of as-forged surfaces unless specific acceptance is obtained from the approving activity.

h. The mechanical drilling of holes in martensitic steels after hardening to strength levels of 180,000 psi and above shall be avoided whenever practicable. When such drilling is unavoidable, detailed information concerning the processes to be used shall be submitted to the approving activity for acceptance.

i. Any necessary straightening of parts after heat treatment to strength levels of 180,000 psi and above shall be accomplished at the tempering temperature, plus O°F (0°C minus 50°F (28°C), or the parts shall receive a stress-relieving treatment at this temperature immediately after straightening. Parts shall be inspected for cracks after straightening.

j. Free-machining grades of steel shall be subject to acceptance by the approving activity.

3.5.1.3.2 Corrosion-resistant Steels. The following limitations shall be observed in the selection and application of corrosion resistant steels.

a. Unstabilized austenitic steels shall not be fusion welded.

b. precipitation hardening semi-austenitic grades shall not be used in applications which require extended exposure to temperatures in the 750°-900°F (399°-482°C) range.

c. The use of heat treated 431 or 19-9DL steel shall be subject to acceptance by the approving activity.

d. The use of 17-4PH stainless steel aged below H1025 shall be subject to acceptance by the approving activity.

3.5.1.3.3 Magnesium Alloy. Magnesium or its alloy shall not be used.

3.5.1.3.4 Castings. Castings shall be in accordance with MIL-C-6021 and shall be classified Class 1B, Grade C or higher. Aluminum alloy castings shall be in accordance with MIL-A-21180. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnation, welding, or other methods acceptable to the procuring activity. Inspection of repaired castings shall be governed by quality control techniques subject to the acceptance by the approving activity,

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3.5.1.3.5 Stress Corrosion Embrittlement Factors. To prevent premature failures caused by stress corrosion or hydrogen embrittlement, the design and method of manufacture of parts of titanium and steel heat-treated to tensile strengths above 220,000 psi or of bare high strength aluminum alloys and the techniques by which they are assembled and installed shall be such that sustained or residual surface tensile stresses and stress concentrations are minimized. Practices, such as: the use of press or shrink fits; taper pins, clevis joints in which tightening of the bolt imposes a bending load on the female lugs; and straightening or assembly operations; which result in sustained or residual surface tensile stresses shall be avoided. In cases where such practices cannot be avoided, corrective practices such as stress relief heat treatment, optimum grain flow orientation, peening (shot or rotary flat) or similar surface working shall be used to minimize the hazard of stress corrosion or hydrogen embrittlement damage. In no case shall sustained or residual surface tensile stresses in these materials exceed 50% of the material specification minimum yield strength in the longitudinal grain flow direction, 35% of the minimum yield strength in the long transverse direction and 15% of the minimum yield strength in the short transverse direction, except that for materials of demonstrated high resistance to stress corrosion cracking, such as 7075-T73 aluminum alloy, such stresses may be as high as 75% of the minimum yield strength in any grain flow direction. These limits for stresses represent the algebraic sum of all the continuous tension and compression surface stresses resulting from any source such as quenching, forming and assembly. The use of cadmium in contact with or plated on titanium shall not be permitted.

3.5.1.3.6 Fatigue Factors. To prevent premature failures caused by repeated loads, the design and method of manufacture, including consideration of the damaging effect of decarburization and certain metallic coatings, of all critical parts and the techniques by which they are assembled and installed shall be such that sustained or residual tensile stresses and stress concentrations are minimized. Practices such as cold straightening, cold forming, and the assembly of mismatched surfaces, which result in sustained or residual surface tensile stresses shall be avoided. In cases where such practices cannot be avoided, corrective practices such as stress relief heat treatment, optimum grain flow orientation, peening (shot or rotary flat) or similar surface working shall be used to minimize premature fatigue failure. In no case shall parts of nonuniform section be cold formed unless the contractor can demonstrate with test results that required fatigue resistance is obtained.

Surface roughness (as defined in ANSI B46.1-1962) and the incidence, location and size of allowable flaws in metallic parts

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subject to repeated loads, wherein stress concentration factors are governing, shall be demonstratably consistent with the design fatigue life of the parts in the planned operating environment. Surface roughness limits and limiting criteria for allowable flaws shall be established for approval by the verifying activity. Alternately, the surface roughness shall not exceed 125 RMS and there shall be no flaws such as tool marks deeper than 0.0002 inch. Such limits for parts critical in low cycle fatigue shall be separately identified.

3.5.1.3.7 Heat Treatment. Heat treatment of aluminum alloys shall be in accordance with the material specification and MIL-H-6088. Steels shall be heat treated in accordance with the material specification and MIL-H-6875. Precautions shall be taken to minimize warpage during heat treatment. Steel parts which require straightening after hardening to 200,000 psi or below may be cold straightened provided a stress relieving heat treatment is subsequently applied.

Straightening of parts hardened to tensile strengths above 200,000 psi shall be accomplished at a temperature within the range from the tempering temperature to 50°F (28°C) below tempering temperature.

3.5.1.3.8 Welding. Welding joints shall be in accordance with applicable welding specifications. Welded assemblies involving steels and section sizes which transform on air cooling to microstructure other than martensite shall be normalized or otherwise processed to equivalent hardness in the weld areas. When welding air hardening steels, appropriate precautionary measures, including preheating interpass temperature control and post heating shall be applied.

To avoid the possibility of stress corrosion and fatigue damage in welded primary structural components; the structures shall be so designed that the weld bead material is accessible for machining during manufacture. In addition, welded primary structural components shall be stress relieved after all welding is completed.

3.5. 1.3.9 Corrosion Protection. Metals which do not possess adequate corrosion-resisting characteristics shall be protected to resist corrosion from dissimilar metal combinations; moisture, salt spray and high temperature deterioration as applicable. The protection of the equipment from corrosion shall be in accordance with MIL-F-7179. Dissimilar metals in immediate contact shall be avoided. Where use of metals in immediate contact cannot be avoided, they shall be insulated. This has particular reference to dissimilar metal contacts defined in MIL-STD-889.

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All metal joints and other contact areas shall be designed or protected to prevent fretting corrosion damage of affected areas.

The following factors will be used in the evaluation of the resistance of the equipment to corrosion. Any corrosion that causes the following shall be deemed sufficient cause for rejection.

a. Malfunctioning of the equipment

b. Shortening of life

c. Impairment of use

d. Impairment of ease of replacement of parts

3.5.1.4 Finish.

3.5. 1.4.1 Aluminum Alloy Parts. Unless otherwise specified by the verifying activity, aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625. In the absence of abrasive conditions and with the approval of the verifying activity, they may be coated with chemical film in accordance with MIL-C-5541. The film deposited by this treatment shall be removed from contact areas required to-act as a path for electrical current.

3.5.2 Parts.

3.5.2.1 Standard Parts. Military standard parts shall be used whenever they are suitable for the purpose, and shall be identified by their military part number. All parts of the same part number shall be interchangeable as applied within the system. Commercial utilitary parts such as screws, bolts, nuts, and cotter pins may be used provided they possess suitable properties and are replaceable by the standard parts (MS and AN) without alteration and provided the corresponding standard part numbers are referenced in the parts list and on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.5.2.2 Threaded Parts.

3.5.2.2.1 Straight Screw Threads. Screw threads shall conform to the requirements of MIL-S-8879 except for electrical connections, fluid fittings and ground and cut threads

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incorporated in proprietary parts which do not have equivalent standards. All exceptions shall conform to MIL-S-7742.

3.5.2.2.2 Tapered Pipe Threads. Tapered pipe threads shall be in accordance with MIL-P-7105 and may be used only for permanently plugging drilled or cored openings.

3.5.2.2.3 Threaded Inserts. All threaded connections in nonferrous materials shall have steel inserts which are suitably protected from electrolytic corrosion.

3.5.2.2.4 Safetying. Provisions shall be made to prevent all screws and screw parts from accidental loosening. Such parts shall be safety wired or fitted with safety stop nuts, where applicable. All other connections liable to be loosened by vibration shall be safetied in a similar manner. If safetying is by wiring, it shall be in accordance with MS-33540. The use of lock nuts which incorporate nonmetallic inserts as a locking feature shall not be permitted.

3.5.2.2.5 Packings. All packings shall be in accordance with the requirements of MS-33666 or MS-33668.

3.5.3 Snapring Retainer. The use of snapring retainers is Prohibited unless failure of the retainer will not affect operation adversely or allow retainer fragments to enter the air passages.

3.6 General Design and Construction Requirement. The starter shall conform to the requirements of this specification.

3.6.1 Acceptable Design. The starter/APU shall be suitable for operational use as an engine starter and auxiliary power unit for supplying emergency electrical power utilizing aircraft fuel in accordance with specification MIL-T-5624, Grades JP-4 and JP-5, and battery for initiation of operation, and its design shall be amenable to production processing methods. Design of each starter/APU shall be consistent with good aircraft accessory practices. Careful consideration shall be given in the design of the starter/APU to simplify servicing, inspection (testing), and overhaul. The necessity of using special tools and fixtures for maintenance and overhaul shall be kept to a minimum.

3.6.2 Interchangeability. Each article of the same part number manufactured in accordance with this specification shall be physically and functionally interchangeable with articles bearing the same contractor part number.

3.6.3 Components. The combination aircraft engine gas turbine starter/APU shall consist of five separate components: a gas

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turbine engine including electrical starter and a generator; a sequence and safety control unit; an ignition unit including the ignition lead; a generator control unit; and a start contactor. The gas turbine engine shall consist of a gas generator section and a power output section, The gas generator will provide hot, high pressure gas to drive the power output section and will incorporate an accessory gear train to drive necessary accessories for operation of starter/APU and to be driven by an electric starter motor to initiate gas generator operation. The power output section through a power turbine and reduction gearbox shall be capable of driving the aircraft engine for starting purposes; be capable of driving a 6KVA electrical generator during APU operation without driving the engine; and be capable of going from APU mode to start mode without shutting down.

3.6.4 Envelope. The gas turbine engine shall be contained within the envelope shown in Figures 1 thru 9. The unit shall not exceed this envelope when hot.

3.6.5 Mounting Provisions. The gas turbine engine shall mount on an accessory drive conforming to U.K. Standard RS623 with 7.200" O.D. V-flange which is suitable for a V-groove, quick attach/detach clamp supplied with the aircraft engine. Adaptor and drive details are shown in Figure 9. Direction of rotation (viewed from the anti-output shaft end) is clockwise.

3.6.6 Weight. The portion of the starter/APU supported by the engine drive pad shall not exceed 129 lbs. dry weight nor shall the resulting overhung moment exceed 970 in-lbs. The sequence and safety control unit weight shall not exceed 4.5 lbs. The high energy ignition unit weight shall not exceed 2.6 lbs. The start contactor weight shall not exceed 2.6 lbs.

3.6.7 Containment. The starter shall provide complete containment of all fragments and shall remain on its mount should failure occur under any operating condition. Failures included are those that occur due to uncontrolled acceleration of the gas generator and failure of the overrunning clutch of the output drive assembly. The containment shall also, in the event of failures, prevent fire external to the starter and external surface temperatures from exceeding 500°F.

3.6.8 Overspeed Protection. The starter/APU shall incorporate a means to prevent the overspeeding of any rotating component resulting from a "no load" or any other condition. The overspeed device shall safely limit the speed of the rotating parts and cut off the fuel supply. If this overspeed protection is remotely mounted any loss of electrical power or signal shall cause the cutoff of the fuel supply.

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3.6.9 Running Engagement. The starter/AP shall suffer no detrimental effects when subjected to running engagements at any output speed up to starter cutoff speed.

3.6.10 External Surface Temperature. The external surface temperature of the starter/APU shall not exceed 500°F during or following any required operating condition. Provisions necessary to meet this requirement shall be integral with the starter and shall not require detachable insulating covers.

3.6.11 Leakage. There shall be no leakage of fluids or gases from the starter/APU other than gases from air exhaust and air inlet during or following operation. The allowable oil leakage at the output drive shall not exceed 10cc per hour.

3.6.12 Drainage. The starter/APU shall incorporate provisions to drain all liquids from air intake, exhaust duct and combustor to prevent accumulations that would adversely affect operation. In addition, fuel manifolds and seals may be drained as required. All drainage shall be collected and piped in accordance with Figures 1 thru 4.

3.6.13 Handling Supports. The starter/APU shall incorporate provisions for carrying, cradling, hoisting and resting the unit on a level surface without damaging lines, cable, fittings, or components attached to the surface of the unit.

3.6.14 Accessibility. Parts of the starter/APU that require routine service checking or replacement shall be readily accessible for servicing and replacement without disassembly of the unit or removal of any major part, component, or accessory.

3.6.15 Environmental Requirements.

3.6.15.1 Temperature. The starter/APU shall withstand and/or operate in the following temperature environments:

a. Engine start -15°F (-26°C) and 122°F (50°C)

b. APU operation -15°F (-26°C) to 122°F (50°C)

c. Overrunning operation from -15°F (-26°C) to 200°F (93°C)

d. Not operating from -65°F (-54°C) to 200°F (93°C)

3.6.15.2 Altitude. The starter/APU shall meet the starter performance requirements herein from sea level to 8000 feet, the APU requirements herein from sea level to 31,000 feet and shall satisfactorily withstand overrunning from sea level to 50,000 ft.

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3.6.15.3 Humidity. The starter/APU shall operate satisfactorily and suffer no detrimental effects from operation in and exposure to relative humidity conditions from 0 to 100 percent including conditions where moisture has condensed on the unit.

3.6.15.4 Salt Spray. The starter/APU shall operate satisfactorily and suffer no detrimental effects from operation in and exposure to an atmosphere of salt laden moisture.

3.6.15.5 Sand and Dust. The starter/APU shall operate satisfactorily and suffer no detrimental effects from limited operation in sand and dust laden air.

3.6.15.6 Fungus. The starter/APU shall suffer no detrimental effects from exposure to fungus growth as may be encountered in tropical climates.

3.6.15.7 Attitude.

3.6.15.7.1 Operational Attitude. The starter/APU shall meet all performance requirements herein when operating with the units centerline tilted from 0° to 13.5° nose up combined with rotation from 5° clockwise to 5° counterclockwise from its normal position. These attitudes apply as the unit is viewed from the anti-drive end.

3.6.15.7.2 Overrunning Attitude. The starter/APU shall show no detrimental effects from continuous overrunning in a 0-10° nose up as well as intermittent overrunning in attitudes from 90° nose up to 90° nose down; and 60 seconds under negative one “g” acceleration load.

3.6.15.8 Vibration.

3.6.15.8.1 Operational Vibration. The starter/APU shall be free of destructive vibration at steady state speeds throughout the operating envelope of the unit.

3.6.15.8.2 Overrunning Vibration. The starter/APU shall be designed to withstand, without damage the vibration environment of Figure 10 regardless of applied direction.

3.6.15.8.3 Remotely Mounted Electrical Controls Vibration. The remotely mounted electrical controls shall be designed to withstand, without damage, the vibration environment of Figure 11 regardless of applied direction.

3.6.15.9 Shock. The starter/APU including all controls and accessories shall be designed to withstand impact shocks in opposite directions along each of three mutually perpendicular

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axes. The shock wave form shall consist of ramp input with a peak amplitude of 20 g's and a nominal duration of 11 milliseconds.

3.6.15.10 Flight and Ground Loading Conditions.

3.6.15.10.1 Design Loads. The starter/APU shall have sufficient strength to withstand limit, ultimate, crash safety strength, operating loads and repeated loads when normally mounted in the aircraft.

3.6.15.10.2 Limit Loads. The starter/APU, when normally mounted in the aircraft, shall meet the performance requirements of this specification after application of the limit load factors shown in Figure 12. There shall be no permanent set after application of limit loads.

3.6.15.10.3 Ultimate Loads. The starter/APU, when normally mounted in the aircraft, shall withstand ultimate loads of 1.5 times the limit loads derived from Figure 12. Distortion and permanent set are permitted; however, there shall be no structural failure. Starter operation is not required after application of ultimate loads.

3.6.15.10.4 Starter Operation Loads. Loads caused by operation of the starter/APU shall be considered in the design. A minimum factor of safety shall be maintained.

3.6.15.10.5 Repeated Loads. The starter/APU shall be capable of meeting the performance requirements of this specification throughout the operational service life of the unit.

3.6.16 Inlet Air and Exhaust Systems. The starter/APU shall provide its own inlet and exhaust systems in accordance with Figures 1 thru 4.

3.6.17 Cockpit Control of Starter/APU. The starter/APU shall be designed such that it can be initiated and operated automatically with not more than two switches. One switch shall control all main engine starts while the other will control the APU operation.

3.6.17.1 Manual Stopping/Abort Device. By opening the cockpit switches of 3.6.17, it shall be possible to terminate the starter/APU operation at any time without damage to the starter/APU.

3.6.18 Cutoff Device. The starter/APU shall be provided with a speed responsive cutoff device which will shut down the unit when the output drive speed reaches 3200 ± 75 rpm in the start mode.

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If primary and secondary cutoff devices are used, they shall both actuate within the same speed limit band.

3.6.19 No Load Operation. In the start mode, the starter/APU shall terminate its operation without damage when the output shaft is unloaded.

3.6.20 Engaging Clutch. The output drive system of the starter/APU shall be provided with suitable clutch mechanisms to automatically engage and disengage the starter/APU during the engine start mode and to prevent power transfer to or from the engine during APU operation. These clutches shall be disengaged whenever the engine drive speed exceeds the starter/APU output speed or the starter/APU is in the APU mode of operation. It shall be possible to translate from APU mode of operation to the start mode of operation without shutting the starter/APU down. Protection shall be provided to insure that no damage will occur to the starter/APU output drive if an operator should inadvertently go from the start mode of operation to the APU mode of operation.

3.6.21 Fuel System.

3.6.21.1 Fuel. The starter/APU shall meet the performance requirements of this specification when using MIL-T-5624 fuel, Grades JP-4 and JP-5 provided the JP-5 fuel is supplied at a temperature not less than that which results in a fuel viscosity of 12 centistokes.

3.6.21.2 Fuel Pressure and Temperature. The starter/APU shall meet the performance requirements of this specification when furnished fuel from the aircraft fuel system at pressures from 0 to 35 psig and temperatures from -30°F to 130°F. (See exception for JP-5 fuel stated in 3.6.21.1).

3.6.21.3 Fuel Filters. Fuel filters, if required, shall have disposable type elements. The elements shall not require replacement before 60 start cycles and/or 10 hours APU operation when using contaminated fuel of 3.6.21.4. Filter element removal and replacement shall not require disassembly of the starter unit. The filter shall provide for bypass of the filter in the event of the filter becoming clogged.

3.6.21.4 Fuel Contamination. The starter shall function satisfactorily when using fuel contaminated to the extent specified in Table I.

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TABLE I - FUEL CONTAMINANTS

CONTAMINANT	PARTICLE SIZE	QUANTITY
Iron oxide	0-5 microns	28.5 gm/1000 gal.
Iron oxide	5-10 microns	1.5 gm/1000 gal.
Sharp silica sand	300-420 microns	1.0 gm/1000 gal.
Sharp silica sand	150-300 microns	1.0 gm/1000 gal.
Preparded dirt conforming to AC Spark Plus Company P/N 1543637 (Coarse Arizona road dust)	Mixture as follows: 0-5 microns (12%) 5-10 microns (12%) 10-20 microns (14%) 20-40 microns (23%) 40-80 microns (30%) 80-200 microns (9%)	8.0 gm/1000 gal.
Cotton Linters	Grade 6, staple below 7; second cut linters. (Per U.S. Department of Agriculture Standards)	0.1 gm/1000 gal.
Crude naphthenic acid		0.03 percent by volume 0.01 percent entrained

Salt water shall be prepared by dissolving 4 parts by weight of salt in 96 parts by weight of distilled or other water containing not more than 200 parts per million of total solids.

3.6.22 Lubrication System. The lubrication system can be provided with oil from the aircraft main engine at a maximum continuous flow rate of 85 gal/hr. with an initial bulk temperature of 300°F in an outside ambient of 122°F and an engine compartment temperature of 158°F. The lubrication system shall be designed to use MIL-L-23699 oil and shall be compatible with MIL-L-7808 oil and its intermittent use.

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3.6.22.1 Oil Level. The starter/APU shall incorporate a suitable system to maintain proper oil levels and prevent oil flooding since the aircraft main engine oil tank will be pressurized up to 5 psi above ambient pressure when the aircraft main engine is operating.

3.6.22.2 Oil Filters. The starter/APU shall provide for the filtration of inlet oil and discharge oil. The discharge oil filter shall be of a 50 micron size to prevent contamination of the aircraft main engine oil in the event of a starter/APU failure. Removal and replacement of filter elements shall not require the disassembly of the starter/APU.

3.6.22.3 Oil Consumption. The oil consumption rate of the starter/APU shall not exceed 60cc per hour.

3.6.22.4 Oil Drain Provisions. Gravity oil drains shall be provided. Magnetic oil drain plugs shall be provided at each drain. The starter/APU shall be designed such that oil draining shall not require disassembly.

3.6.23 Starter Initiator System. The starter initiator shall be an electric motor designed to operate from 24 volt dc 15 ampere hour sealed lead acid battery military Part No. D8565/3-1 and 28 volt dc power conforming to MIL-STD-704 capable of producing a minimum of 500 amperes.

3.6.23.1 Initiator Engaging Clutch. The initiator shall be provided with a clutch which shall automatically engage and disengage the initiator from the starter accessory drive.

3.6.23.2 Initiator Running Engagement. The starter shall suffer no detrimental effects when subjected to running engagements of the initiator motor.

3.6.23.3 Initiator Torque Limiting. The starter initiator motor shall incorporate, if required, a positive type torque limiting device to protect the starter from excessive shock loads during engagement or re-engagement. Initiators utilizing output drive shear sections must be designed to permit replacement of the shear section without disassembly of the initiator or accessory gearbox.

3.6.24 Electrical Requirements.

3.6.24.1 Starter/APU Initiator. The starter/APU shall be successfully started and operated by either a 15 ampere hour sealed lead acid battery military Part No. D8565/3-1 or a 28 volt dc external power source conforming to MIL-STD-704 capable of producing a minimum of 500 amperes.

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3.6.24.2 Starter/APU Output Generator. The starter/APU power Producing generator shall have a three phase star connected stator winding with all three phases and the neutral brought out. The output shall be a nominal 201 volts RMS line to line, within a frequency range of 380 to 420 Hz. The output shall meet the electrical characteristics of MIL-STD-704.

3.6.24.3 Connectors. External electric connectors shall be in accordance with MIL-C-38999. Provisions for lockwiring the connector plugs shall be incorporated at the electrical receptacle.

3.6.24.4 Internal Electric Connections. Internal connections shall be made in accordance with MIL-E-5400 and MIL-STD-454.

3.6.24.5 Explosion-Proof. Electrical system components shall be of the explosion proof type, and shall not constitute a fire nor explosion hazard to the aircraft when installed in an area where flammable or explosive vapors may be present.

3.6.24.6 Electromagnetic Interference (EMI). The safety and control unit shall be designed to meet the electro-magnetic compatibility requirements of MIL-STD-461, Notice 3, Class F which includes tests CE03, RE02, and RS03. However, the field intensity for test RS03 shall be as follows:

20 volts/meter from 14 KHz to 200 MHz
65 volts/meter from 200 MHz to 450 MHz
20 volts/meter from 450 MHz to 1.0 GHz
100 volts/meter from 1.0 GHz to 40.0 GHz

3.6.24.7 Dielectric Strength. The starters electrical system and system components shall be designed to meet the requirements of MIL-E-25499.

3.6.25 Provisions For Maintainability. Built in test equipment, construction and packaging, and provisions for test points shall be as specified in specification AR-10.

3.6.25.1 Compatibility with VAST. Equipment shall be compatible with ATE (Automatic Test Equipment) and shall meet the requirements of MIL-STD-2076. ATE test program shall be furnished in accordance with MIL-STD-2077. If changes are made which will affect the fault diagnosis procedure, changes to the existing test program shall be prepared as part of the equipment changes in accordance with MIL-STD-2077.

3.7 Performance Requirements.

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3.7.1 Starter Requirements.

307.1.1 Starter Output Torque. The transient and steady-state output torque of the starter/APU when operating in the start mode with the required inlet and exhaust ducts installed shall meet the requirements of Figures 13 and 14 for the ambient temperature of -15°F (-26°C) to 122°F (50°C) at sea level and -15°F (-26°F) to 95°F (35°C) at 8,000 ft. altitude.

3.7.1.2 Starter Duty Cycle. When operating on the ground within the environmental conditions of altitude from sea level to 8,000 ft. and inlet temperatures the starter/APU shall be capable of performing three consecutive 40 second start or cranking cycles. A 20-minute cooling period may be imposed before three more start or crank cycles can be attempted. The starter/APU service life shall not be degraded by this duty cycle.

3.7.2 APU Requirements.

3.7.2.1 Ground Operation. The starter/APU shall be capable of starting and accelerating the unit to operating speed in the APU mode and to provide up to 6KVA of electrical power within an altitude range of sea level to 8,000 ft. and within an ambient temperature range of -15°F (-26°C) to 122°F (50°C).

3.7.2.2 Flight Operation. The starter/APU shall be capable of starting and accelerating the unit to operating speed in the altitude range of sea level to 20,000 ft. and of providing up to 6KVA of electrical power up to 31,000 ft.

3.7.2.3 Translation. When the starter/APU is operating in the APU mode - on the ground - it shall be possible to make an immediate and automatic translation from the APU mode to start mode without requiring a shutdown of the unit.

3.7.3 Operating Service Time. The starter/APU shall be capable of operating for a minimum of 1250 starts or translations, 450 hours APU operation and 1000 hours of overrunning without maintenance. The starter/APU performance deterioration from initial calibration to the completion of the minimum service life shall not be greater than 5 percent.

3.8 Installation.

3.8.1 Installation Drawing. The supplier’s installation drawing shall be accepted by the approving activity and will form a part of the first article test report. Manufacturers supplying a starter/APU to this specification shall coordinate installation with the approving activity prior to fabrication of verification units.

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3.9 Reliability and Maintainability.

3.9.1 Reliability Program. The supplier shall prepare a proposed reliability program plan for the starter/APU in accordance with MIL-STD-785 and shall submit it to the approving activity. First article approval will not be granted to a product until a reliability program for the product has been approved by the approving activity, in addition to the product's passing the first article tests specified herein.

3.9.1.1 Reliability Target. The target Mean Time Between Failures (MTBF) for the starter shall be 1000 hours. This MTBF is based on the endurance test cycle defined in section 4.5 of this specification.

3.9.2 Maintainability Program. The supplier shall prepare a proposed maintainability program plan for the starter in accordance with MIL-STD-470 and shall submit it to the approving activity. First article approval will not be granted to a product until a maintainability program for the product has been approved by the approving activity, in addition to the product's passing the first article tests specified herein.

3.9.2.1 Quantitative Requirements.

3.9.2. 1.1 Direct Maintenance Manhours/Flight Hour (DMMH/FH). The starter/APU shall be designed such that DMMH/FH shall not exceed 0.0699.

3.9.2.1.2 Mean-Time-To-Repair (MTTR). The MTTR for organizational level unscheduled maintenance shall not exceed 1.7 elapsed hours, excluding aircraft access. Repair at this level shall include troubleshooting, fault isolation and functional checkout, utilizing BIT and built-in failure monitoring features where designed into the equipment. Organizational level maintenance shall be accomplished without use of Peculiar Ground Support Equipment (PGSE).

3.10 Marking.

3.10.1 Identification of Product. Equipment, assemblies, and parts shall be marked as specified in Standard MIL-STD-130. Starter/APU shall be marked with the following information:

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STARTER/APU, AIRCRAFT ENGINE, GAS TURBINE,
MODEL_____
Specification: MIL-S-85419
Manufacturer's part number_____ **LESS ISSUE_____
Manufactuer's serial number_____
Government order or contract number_____
National stock number_____

NOTES : ** For use by the Naval Air Rework Facilities

3.10.2 Marking of Product. The military part number shall not be used on a starter/APU until notification has been received in writing from the approving activity that the starter/APU has been accepted for use on Navy aircraft.

3.10.3 Marking of Parts. Each part and assembly shall be marked with its part number and drawing revision symbol where applicable, which shall be the same as the drawing number, except the following:

3. 10.3.1 Those parts which do not have a suitable or sufficient surface for a part number,

3.10.3.2 Those parts which are permanently assembled by welding, brazing, soldering, or riveting shall carry the assembly part number.

3.10.4 Part Number. The part number shall, when practicable, be located to permit its being read after assembly of the complete unit. The marking shall be such that it will not be affected as a result of service usage during the life of the part.

3.11 Part Number and Parts List

3.11.1 Part Numbers. Starter part numbers are not to be coded to describe Part characteristics and shall be limited to a maximum of 10 digits or characters. All parts shall be marked with the latest drawing revision letter or number to which the part was made.

3.11.2 Parts List. A parts list is a tabulation of the items necessary to fabricate or assemble the starter to which the list applies. The parts list for the starter which is granted first article approval based on (1) the satisfactory completion of the first article tests and (2) the approval of the first article test detail inspection and test report by the approving activity shall constitute the approved parts list for any subsequent starter of the same model to be delivered to the Government. All parts shall be marked directly with an identification number and the vendor or fabrication source of that particular part, unless

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otherwise specified in the parts list. The contractor shall assign an item configuration control number to the starter configured by the approved parts list, and this number together with the approved parts list shall be continuously updated to reflect the incorporation of Class I or Class II engineering changes accepted by the approving activity. This item configuration control number will be used to identify the starter to be delivered and will also be used for final acceptance. The approved parts list shall be prepared in accordance with Chapter 3 of DOD-STD-100 and this list shall also show the latest drawing revision letter or number used to manufacture the part incorporated in the starter at the time of acceptance. The use of the so-called “clip system” of attaching engineering change forms to the applicable drawing in lieu of showing the change(s) in the body of the drawing shall not be permitted.

3.11.3 Design and Construction Changes.

3.11.3.1 Changes in Design. No changes shall be made in the design or materials of a starter/APU that has been approved in accordance with this specification, except when the changes are accepted by the approving activity in accordance with DOD-STD-480.

3.11.3.2 Changes in Part Numbers. Part number changes shall be controlled in accordance with 1-302.14 of DOD-STD-100.

3.11.3.3 Changes in the Parts List. Changes to the approved parts list shall be governed by the requirements specified in 3.11.3.2.

3.11.3.4 Changes in Vendors. Changes in a vendor or fabrication source of those parts used in the first article tests shall be in accordance with the following procedure: the contractor shall prepare and submit to the approving activity a list of tests to approve an alternative vendor source. The specific tests required to substantiate an alternate vendor as a supplier of a starter part shall be defined. The fabrication source of selected vendor components will be included in this list. The list shall be subject to acceptance by the approving activity. The contractor thereafter shall be responsible for insuring that all parts, components, and assemblies on the substantiation list comply with the approved fabrication source, and that any changes to those sources are effectively controlled. The contractor shall be responsible for performance of substantiation tests to establish satisfactory alternate vendors for fabrication sources. A fabrication source is defined as the prime physical source producing the part, component, or assembly. Changes of fabrication location, such as to another plant of an individual vendor, shall be construed as a change of fabrication source.

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3.11.3.5 Responsibility for Changes. Acceptance by the approving activity of changes to the first article test starter does not relieve the manufacturer of full responsibility for the results of such changes on any starter characteristic.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Inspection. Inspection and testing shall be classified as follows and as listed in Table II.

a. First Article Inspection. First Article tests are those tests which verify that equipment being supplied demonstrates compliance with the requirements of this specification. The right is reserved by NAVAIRSYSCOM to require any additional tests deemed necessary to determine compliance with this specification.

b. Quality Conformance Inspection.

(1) Acceptance Test. Acceptance tests are those tests accomplished on each equipment manufactured and submitted for acceptance under a production contract.

(2) Sampling Tests. Sampling tests are those tests conducted on production samples to assure that production equipment being delivered are identical to approved units and meet all the requirements of this specification.

4.2 First Article. Equipment will be considered acceptable if the verification samples successfully complete the first article tests of this specification as designated by Table II and upon acceptance by the approving activity of the maintenance and reliability plan.

4.2.1 Verification Test Sample. The first article test sample shall consist of (a) two starter/APUs which have successfully completed the acceptance tests by the contractor; (b) a mating plug for each electrical connector, used on each unit; (c) the test report described in 4.2.6; and (d) two sets of component wiring diagrams, outline drawings, detail assembly drawings with

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parts list, and operating instructions. The first article test samples shall be plainly identified by securely attached tags clearly and durably marked with the following information, and shall be forwarded to the Commanding Officer, Naval Air Propulsion Center, Trenton, New Jersey 08628, after obtaining authorization from the approving activity for submittal of first article test samples.

TABLE II - TEST PLANS

TEST PARAGRAPH NUMBER	TEST PARAGRAPH TITLE	FIRST ARTICLE UNIT NUMBER 1	FIRST ARTICLE UNIT NUMBER 2	ACCEPTANCE
4.5.1	Calibration	X	X	
4.5.2	Endurance	X		
4.5.3	Surface Temperature	X		
4.5.4	Leakage	X		
4.5.5	Shock Tests		X	
4.5.6	Vibration		X	
4.5.7	Start Cycle Abort	X		
4.5.8	Running Engagement	X		
4.5.9	Teardown Inspection	X		
4.5.10	Containment	X		
4.5.11	Initiator Running Engagement		X	
4.5.12	Initiator Torque Limiting	X		
4.5.13	Initiator Engaging Clutch		X	
4.5.14	Humidity		X	
4.5.15	Salt Spray		X	
4.5.16	Sand and Dust		X	
4.5.17	Fungus		X	
4.5.18	Fuel Contamination		X	
4.5.19	Flight and Ground Loads		X	
4.5.20.1	Explosion Atmosphere		X	
4.5.20.2	Electromagnetic Interference		X	
4.5.20.3	Dielectric Strength		X	
4.5.21.2	Examination of Product			X
4.5.21.3	High Speed Components			X
4.5.21.4	Leakage			X
4.5.21.5	Performance			X
4.5.21.6	Dielectric Strength			X
4.5.21.7	Burn-in			X

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Sample for First Article Test
Submittal by (name) (date) for first article test in accordance with the requirements of
Specification (date) under authorization
(reference letter authorizing tests).

Starter/Auxiliary Power Unit; Aircraft Engine. Gas Turbine Type
Manufacturer's Part Number
Name of Manufacturer
Navy Work Unit Assignment Number

4.2.2 First Article Testing. First Article testing shall be specifically authorized by the approving activity. At the option of the approving activity, any or all first article tests required may be conducted on starter/APUs submitted for first article, for sampling, for production acceptance, or on any unit delivered under a production contract for which conformance to this specification is required.

4.2.3 Sampling Tests. Each equipment of the same part number delivered as an approved item under this specification shall be numbered sequentially in essentially the order that it is submitted for government acceptance regardless of the contract under which delivered or by whom purchased. To meet the delivery requirement for the first 50 components, 51 components must successfully complete the required acceptance tests; 50 of which are to be shipped as directed by the applicable contract(s); and one component selected at random by the Government Inspector is to be forwarded to the Naval Air Propulsion Center (PE61), Trenton, New Jersey 08628. Additional samples will not be required unless directed by the applicable contract. The span of serial numbers of the Production units from which each sample was selected shall be recorded and this information included with the sample. The sample shall be shipped no later than the final unit in the production group from which selected. The units forwarded to the Naval Air Propulsion Center will be inspected and examined for conformance to this specification. Tests will be conducted at the discretion of the approving activity. The sample unit will be returned to the supplier with a report of the government test results at the conclusion of the sampling tests. Delivery of production units successfully passing the acceptance tests is to be accepted until supplier is officially notified of noncompliance of units with this specification.

Shipping authorization and instructions are to be given by the cognizant Government Inspector when the sample unit is selected.

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4.2.4 Rejection and Retest First Article Samples. Samples which have been rejected or returned to the manufacturer for any reason during first-article tests, may be reworked or have parts replaced to correct defects. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the supplier must be submitted in writing by the supplier to the approving activity. Tests shall not be resumed until such a report is received. Where first article tests are conducted under the auspices of the supplier, the approving activity will be advised upon failure of a first article sample and of the action taken by the supplier with regard to failure. If the supplier does not correct and return components rejected during testing within sixty days after he receives them from the approving activity or does not submit a program acceptable to the approving activity for the correction of deficiencies noted within thirty days, the component will be considered unacceptable.

4.2.5 Disassembly and Inspection. At the conclusion of tests, each sample will be disassembled as necessary to inspect for wear, defects and overheating.

4.2.6 Test Report. A test report is required and shall be prepared in accordance with MIL-STD-831.

4.3 Acceptance.

4.3.1 Acceptance Requirements. Each unit produced under this specification will be accepted if:

a. Conforms to this specification and if the conformance is sustained in accordance with paragraph 4.2.3.

b. Acceptance tests as designated by Table I are successfully completed.

c. All changes to equipment have been processed in accordance with MIL-STD-480.

4.3.2 Rejection and Retest of Units Submitted For Acceptance. Units that have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government Inspector.

4.4 Test Conditions. Unless otherwise specified in this specification, all tests shall be conducted at an ambient temperature range of 50° - 120°F (10° - 49°C) and with the

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exhaust conditions at the prevailing laboratory atmospheric pressure.

4.4.1 Test Apparatus.

4.4.1.1 Accuracy. For all starter/APU and component tests, the test apparatus shall be such as to insure that the reported data will have a steady state accuracy of ±1 percent of the maximum rated value, except that temperature shall be accuracte to ±5°F and weight shall be accurate to ±0.2 percent.

4.4.1.2 Flywheel. For verification and acceptance testing, a flywheel shall be used to load the output shaft. The flywheel shall have a polar moment of inertia of 200 $lb\text{-}ft^2$ (8.43 $kg\text{-}m^2$).

4.4.2 Instrumentation. Instrumentation shall be provided during verification and acceptance tests to observe or determine the following data as minimum:

a. Output shaft speed
b. Output shaft torque
c. Inlet air temperature
d. Inlet air pressure
e. Exhaust gas total temperature
* f. Exhaust gas total pressure
* g. External surface temperature (4 locations)
h. Fuel inlet pressure
* i. Fuel flow rate
* j. Oil consumption
* k. Oil temperature(s)
l. Gas generator speed
m. Starter voltage
n. Starter current
o. Generator voltage
p. Generator current
q. Generator frequency

*NOTE: These items need not be instrumented during the acceptance test unless specifically required to determine acceptance performance.

4.4.2.1 Recording Equipment. Automatic recording equipment shall be used during those tests regarding evaluation of time versus starter or component variable.

4.5 Tests. The tests are complete when the starter/APU is operating satisfactorily at the end of the tests herein, recalibration does not reveal excessive performance deterioration and teardown inspections do not disclose parts failures or

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indicate imminent failure which might compromise safety of flight.

4.5.1 Calibration.

4.5.1.1 Starter Mode. The starter/APU shall be subjected to three start cycles against the flywheel at each of the conditions specified in Table III and the performance shall meet or exceed the requirements of Figures 13 and 14.

TABLE III - CALIBRATION TEST CONDITIONS, STARTER MODE

TEMPERATURE			ALTITUDE
Starter Ambient	Inlet Air	Outlet Fuel	
-15°F(-26°C)	-15°F	-15°F	sea level*
+59°F(+15°C)	+59°F	+59°F	sea level*
+130°F(+54°C)	+130°F	+130°F	sea level*
-15°F(+26°C)	-15°F	-15°F	8000 feet
+95°F(+35°C)	-95°F	-95°F	8000 feet

*May be conducted at local prevailing ambient provided that data are corrected to sea level.

4.5.1.2 APU Mode. The starter/APU shall be started and operated as an APU while performance is determined at the conditions specified in Table IV.

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TABLE IV - CALIBRATION TEST CONDITIONS, APU MODE

Inlet Temperature	Altitude	Mach No	Load Range
-15°F(-26°C)	sea level	0	0-6KW
+59°F(+15°C)	sea level	0	0-6KW
+122°F(+50°C)	sea level	0	0-6KW
-15°F(-26°C)	8,000	0	0-6KW
+95°F(+35°C)	8,000	0	0-6KW
+62°F	sea level	0.16	0-6KW
+97°F	sea level	0.40	0-6KW
+0°F	19,000	0.31	0-6KW
+19°F	19,000	0.55	0-6KW
-20°F	25,000	0.35	3-6KW
+ 2°F	25,000	0.61	3-6KW*
-42°F	31,000	0.40	6KW*
-18°F	31,000	0.67	6KW*

*Starts will not be required at these conditions.

At the conclusion of this endurance test, the performance deterioration shall not be greater than 5 percent.

4.5.2 Endurance. The starter/APU shall be subjected to an endurance test in accordance with the following schedule:

a. 500 hours overrunning operation

b. Endurance Cycle A - Repeated 130 times

(1) 3 Start cycles
(2) 2 Hours APU operation
(3) 1 Translation cycle
(4) 2 Start cycles
(5) 1 Cranking cycle
(6) 1 Hour APU operation
(7) 1 Translation cycle

Endurance Cycle B - To be conducted after every 13th Cycle A endurance

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(1) 2 Cranking cycles
(2) 1 Start cycle
(3) 2 Hours APU Operation
(4) 1 Translation cycle
(5) 2 Cranking cycles
(6) 1 Start cycle
(7) 2 Hours APU operation
(8) 1 Translation cycle
(9) 2 Cranking cycles
(10) 1 Starting cycle
(11) 2 Hours APU operation
(12) 1 Translation cycle
(13) 1 Cranking cycle
(14) 1 Starting cycle
(15) 3 Rapid starting cycles

d. 500 hours overrunning operation

4.5.2.1 Overrunning. The starter/APU shall be subjected to 1000 hours of overrunning with the output shaft driven at 11,000 ± 100 rpm in a clockwise direction (viewed anti-output shaft end) in accordance with Table V. The ambient air temperature shall be maintained at a minimum of 150°F (66°C) and the mounting pad shall be maintained at a temperature of 200°F (93°C).

TABLE V - OVERRUNNING TEST ALTITUDE

Time Minutes	Attitude
4	0°
6	+45°
4	0°
30	0°
4	0°
2	-45°
10	0°

NOTE: Position of the starter anti-drive end with respect to the horizontal centerline. Lubricate addition or change is permitted after 300 endurance test cycles.

4.5.2.2 Starting Cycles. The starter/APU shall be subjected to start cycles at a rate of not less than 6 cycles per hour as specified herein. Each cycle shall consist of an acceleration of the inertia load to a dwell condition following by an acceleration to cut out speed in accordance with Table VI:

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TABLE VI - START CYCLE TEST CONDITIONS

Test Stand Rotor Polar Moment of Inertia	Dwell		Cutout		
	Speed	Torque	Time	Speed	Time
LB-FT²	RPM	LB-FT (MIN)	SEC	RPM (MIN)	
200	1550±50	200	10	3200	10-0+5

NOTE : The above values do not allow for flywheel windage losses, test stand drag, or inertia of the starter. In order to determine compliance with this specification, the test equipment must be calibrated and specific test results corrected accordingly. If the acceleration specified to cutout is not developed because of test stand drag torque, the test stand inertia load may be reduced or external power may be applied to assist acceleration to cutout speed.

4.5.2.3 Rapid Starting Cycles. Ten series of rapid starts shall be run as specified in paragraph 4.5.2.2. Each series shall consist of three start cycles with no more than 60 seconds between completion of one cycle and the initiation of the next. A thirty-minute cooling period will be observed prior to the next starter operation.

4.5.2.4 Cranking. The starter/APU shall be subjected to cranking cycles of 40-second duration with the output shaft speed limited, by a resistive torque, to 1700 to 1900 rpm as specified in paragraph 4.5.2.2.

4.5.2.5 APU Operation. The starter/APU shall be operated in one hour and two hour cycles as specified in paragraph 4.5.2. A one hour cycle shall consist of

a. APU start
b. 5 minutes at 1.5KVA
c. 20 minutes at 6KVA
d. 4 minutes at 3KVA
e. 1 minute at no load
f. 4 minutes at 3KVA
g. 20 minutes at 4.5KVA
h. 6 minutes at 1.5KVA

The two hour cycles will double the time at each condition.

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4.5.2.6 Translation Cycles. The starter/APU shall be subjected to translation from APU mode to main engine starting upon the conclusion of each APU operational cycle as specified in paragraph 4.5.2.

4.5.3 Surface Temperature. During the initial portion of the verification test, a surface temperature survey shall be conducted during cycling, rapid cycling, and cranking at ambient temperature, After determining the hottest areas on the starter/APU exterior surface, these areas will be monitored throughout the test for compliance with 3.6.10. Unless otherwise specified, maximum surface temperature during and after shall be recorded for every twenty fifth cycle during the endurance test.

4.5.4 Leakage. During all verification test operations, checks shall be conducted for leakage of fuel, exhaust gas and lubricant. Evidence shall be obtained to verify compliance with leakage requirements of 3.6.11.

4.5.5 Shock Tests. The starter/APU shall be attached to the test fixture by the mounting means used in the service installation plus all electrical cables and any other connections necessary for proper operation. The starter shall be subjected to three shocks in each direction on each of three mutually perpendicular axes. The shock waveform shall consist of ramp input with a peak amplitude of 20g’s and a nominal duration of 11 milliseconds. The peak amplitude must be within ± 10% of the specified value measured with a 250 Hz low pass filter. The duration must be within ± 1 millisecond. If the electrical controls are separately mounted, they shall be shock tested separately in their service mounting with electrical cables attached.

4.5.6 Vibration. The vibration test shall be conducted in accordance with the method and procedure as follows:

4.5.6.1 Method. The starter/APU shall be tested as a complete assembly. Before and after this test, a calibration shall indicate that no component subject to calibration has changed its calibration beyond the design tolerance range. All components not subject to calibration shall be operated and shall not have had any part of their function impaired by the test.

4.5.6.2 Procedure. The following procedure shall apply:

4.5.6.2.1 Test Installation. The starter/APU shall be mounted on a test fixture to simulate aircraft installed position. The test fixture shall be as rigid as possible and designed to eliminate or minimize fixture resonances in the test frequency range. The test configuration shall be such that rotational

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motion of the fixture is not induced due to any unsymmetrical weight or stiffness distribution of the equipment, Measurements of vibratory accelerations or displacements shall be made at the mounting base of the valve. All frequency measurements shall be accurate within ± 5 percent. All displacement or acceleration measurements shall be accurate within ± 10 percent. The motion of the vibrator table shall be simple harmonic motion with not more than 20 percent distortion.

4.5.6.2.2 Resonance Survey. The equipment shall be vibrated with essentially simple harmonic motion in each of three mutually perpendicular directions to explore for resonant frequencies. Frequency scans between 5 and 1000 Hz shall be made very slowly and carefully. The frequency range shall be broken into small convenient intervals and each interval scanned at a 0.01 inch double amplitude or ± 2g, whichever is less. The fixture motion shall be observed closely during the frequency scans to detect the frequencies of minimum fixture motion which defines the frequencies at which major components of the unit are in resonance, These frequencies are quite sharply tuned and do not necessarily coincide with the frequencies at which maximum component amplitudes or noises occur when scanning at constant applied force amplitude. When the resonant components are small, the reduction of fixture motion at resonance may not be discernible, in which case the resonant frequencies may be determined from observations of maximum component amplitudes or noises, phase shift between fixture motion, and motion of a resonant part. In all cases, the resonant frequencies shall be verified, if possible, by checking for minimum fixture motion. The test report must contain complete descriptions of all resonances, including the resonant frequencies, parts in resonance, and modes of vibration.

4.5.6.2.3 Resonance Endurance Tests. After completion of the resonance survey, a resonance endurance test shall be conducted at all resonant frequencies, with applied force adjusted as necessary to obtain the applicable fixture displacements or accelerations specified in Figure 10. The vibrator and power supply must produce the prescribed fixture motions of all frequencies. Endurance tests shall not be conducted at any frequency at which the fixture amplitude abruptly increases when scanning at a constant applied force amplitude or at which the transmissibility (output/input) is less than two. Resonance endurance tests shall be conducted for a period of 30 minutes at the four most severe resonant points on each axis.

4.5.6.2.4 Cycling Endurance Test. The starter/APU shall be subjected to the following cycling tests at room temperatures in addition to the resonance survey and resonance endurance tests. The starter/APU shall be mounted as specified in 4.5.6.2.1 and

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shall be vibrated at the displacements and accelerations defined by Figure 10 with the frequency continuously varying at a logarithmic rate from 5 to 1000 Hz and back to 5 Hz in approximately 20 minutes at double amplitudes or vibratory acceleration levels of Figure 10. This shall be repeated for a total of five cycles on each axis.

4.5.6.3 Electrical Controls Random Endurance Test. If the electrical controls for the starter/APU are separately mounted they shall be mounted on text fixture the same as it would be in the service installation mounting. The electrical controls shall be continuously excited for three hours along each of three mutually perpendicular axes in accordance with Figure 11. During these tests, the instantaneous random vibration peaks may be limited to 3 times the rms acceleration level. The rms acceleration level shall not be less than the specific minimum value. Resonant modes of the moving mass (vibration exciter moving element, fixture and the test item) shall be equalized or compensated for within the frequency range of the test curve. The applied vibration spectrum shall normally be within the tolerance of ± 1.5 db between the frequencies of 10 to 1000 Hz. A wave analyzer shall be used to assure the specified equalization tolerances. The wave analyzer shall have a maximum bandwidth of 25 Hz below 1000 Hz, with a sweep rate (Hz/second) not exceeding .034 times the filter bandwidth squared, The integration time constant shall not be less than one second.

4.5.7 Start Cycle Abort. A minimum of two start cycle aborts shall be conducted at output shaft speeds of 500, 1000, and 2500 rpm by interrupting the electrical start control signal. One abort shall be completed after each 50 endurance start cycles.

4.5.8 Running Engagement. The starter/APU shall be operated to cutout speed after which it will be subjected to engagements during coast down at speeds ranging from zero to cutout speed in equal increments selected to provide a minimum of ten engagements. No damage to the starter/APU shall result from this test.

4.5.9 Teardown Inspection. A teardown inspection consisting of complete disassembly of the starter/APU shall be made at the completion of the qualification test. Dimensional inspection shall be made of all wearing parts. Rotating parts shall be subjected to Magnaflux of Zyglo inspection, except that assembled anti-friction bearings shall not be subjected to Magnaflux inspection. Rotating parts subjected to centrifugal stresses shall be measured on dimensions likely to show growth. Where practicable, seal clearances shall be determined. Failure of or any abnormal wear or damage to any part, sufficient to impair

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continued use of the starter, shall be sufficient cause for rejection of the starter.

4.5.10 Containment. After completion of the inspection of 4.5.9, the following containment demonstrations shall be required. Unless oherwise specified, the flywheel shall be attached for all cycles. The test shall demonstrate compliance with the containment requirements of 3.6.7. Parts may fall from the starter exhaust port provided they contain no destructive energy. This shall be demonstrated by placing the starter inside a cylindrical shaped witness shield placed within one foot, except for the exhaust area, where the witness shield shall be placed within three feet of the starter exhaust port located such that the exhaust gas stream will impinge normal to the shield. The witness shield shall be constructed of soft aluminum (0.032 inch maximum thickness) conforming to Federal Specification QQ-A-250/1C Aluminum Alloy 1100, Plate and Sheet.

4.5.10.1 Uncontrolled Overspeed Power Output Section. The starter shall be subjected to an uncontrolled overspeed condition (in the power output and speed reduction assemblies) by operating with all cutoff devices in the power output and speed reduction assemblies rendered inoperative and the flywheel removed for a period of not less than 40 seconds.

4.5.10.2 Uncontrolled Overspeed Gas Generator Section. The starter/APU shall be subjected to an uncontrolled overspeed condition (in the gas generator section) by operating with all fuel controls disabled to provide maximum fuel flow and all cutoff devices in the gas generator section rendered inoperative.

4.5.10.3 Overrunning Clutch Failure. The starter output shaft overrunning device shall be rendered inoperative. The output shaft shall be driven to 11,000 rpm and maintained for a minimum of 5 minutes.

4.5.11 Initiator Running Engagement. The starter/APU shall be subjected to five running engagements of the starter’s initiation motor.

4.5.12 Initiator Torque Limiting. If the initiator motor incorporates a shear section, the shear section shall be tested to failure to demonstrate overtorque protection.

4.5.13 Initiator Engaging Clutch. The starter/APU shall be subjected to a maximum of 5 start cycles with the initiator engaging clutch rendered unable to disengage. Failure of the starter initiator motor shall be cause for termination of this test. Failure of the initiator shall demonstrate design conformance with the containment requirements of 3.6.7.

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4.5.14 Humidity. The starter/APU shall be subjected to a humidity test in accordance with MIL-STD-810 Method 507. The starter/APU shall accomplish two start cycles and one hour of APU operation before and after the test and one after each 24-hour humidity cycle. All openings normally left exposed in the aircraft installation shall be left open during the test. Data obtained shall be corrected to show sea level performance on a 59°F (15°C) day.

4.5.15 Salt Spray. The starter/APU shall be subjected to a salt spray test in accordance with MIL-STD-810 Method 509. The starter/APU shall accomplish two start cycles and one hour of APU operation before and after the test and one after, 12, 24, and 36 hours of exposure. All openings normally left exposed in the aircraft installation shall be left open during the test; however, salt spray shall not be caused to impinge directly into the inlet or exhaust openings. Data obtained shall be corrected to show sea level performance on a 59°F (15°C) day.

4.5.16 Sand and Dust.

a. Non-operating - The starter/APU shall be subjected to a sand and dust test in accordance with MIL-STD-810, Method 510. All openings normally open on the aircraft installation shall be left open during the test, however, the sand and dust shall not be caused to impinge directly into the inlet or exhaust ducting. The starter/APU shall be subjected to two start cycles and one hour of APU operation before and after the test.

b. Operating - The starter/APU shall be subjected to four cycles of sand and dust ingestion at ambient temperatures and pressure conditions. The concentration of sand and dust shall be .0015 grams per cubic foot of inlet air. The sand and dust contaminant shall consist of crushed quartz with the total particle size distribution as specified in Table VII:

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TABLE VII - PARTICLE SIZE DISTRIBUTION

Particle Size, Microns	Quantity Percent by Weight Finer than Size Indicated
1,000	100
900	98-99
600	93-97
400	82-86
200	46-50
125	18-22
75	3-7

Data obtained shall be corrected to show sea level performance on a 59°F day, Any performance deterioration during this test shall be noted in the test report.

4.5.17 Fungus. The starter/APU shall be subjected to a fungus test in accordance with MIL-STD-810 Method 508 Procedure I. The starter/APU shall be subjected to two start cycles and one hour of APU operation before and after the test. Data obtained shall be corrected to show sea level performance on a 59°F day. NOTE: This test may be waived if the starter does not contain fungus nutrients and a statement to this effect is included in the verification test report. A list of typical materials which will support and are damaged by fungi is given in MIL-STD-810 Method 508.

4.5.18 Fuel Contamination. The starter/APU shall be subjected to 240 start cycles and/or 100 hours APU operation with contaminated fuel in accordance with 3.6.21.4. Fuel system filter(s), if applicable, shall have sufficient capacity to operate without restricting the required fuel flow for 60 start cycles and/or 10 hours APU operation.

4.5.19 Flight and Ground Loading. Limit and utlimate load tests are required to demonstrate the strength requirements of the equipment and its mounting. The equipment shall be mounted in a manner simulating the aircraft installation. The mounting hardware shall be the same as used in the aircraft and shock mounts shall be in place if normally used.

4.5.19.1 Test Apparatus. A centrifuge capable of producing the required load factors on the equipment shall be used for these tests.

4.5.19.2 Test Procedure. The equipment shall be mounted on the centrifuge so as to produce a force in the required direction.

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The test time duration in each direction shall be at least one minute following centrifuge stabilization.

4.5.19.3 Limit Loads. The equipment shall be subjected to the limit load factors specified in Figure 12. The equipment shall satisfactorily demonstrate its specified performance after the limit load tests. No distortion or permanent set is permitted after application of limit loads.

4.5.19.4 Ultimate Loads. The equipment shall be subjected to the ultimate load factors of 1.5 times the limit loads of Figure 12. Distortion and permanent set are permitted; however, there shall be no structural failure. Equipment operation is not required during or after application of ultimate loads.

4.5.20 Starter/APU Electrical System.

4.5.20.1 Explosion Atmosphere. The starter/APU electrical system shall meet the requirements of 3.6.24.5 when tested in accordance with MIL-STD-810, Method 511.1 Procedure I except that the explosion atmosphere shall be a mixture of air and JP-4 (MIL-T-5624) with MAIT of +436°F (819°C). The test shall be limited to an altitude of 30,000 feet.

4.5.20.2 Electromagnetic Interference. The starter/APU electrical system shall meet the requirements of 3.6.24.6 when tested in accordance with applicable portions of MIL-STD-462.

4.5.20.3 Dielectric Strength. The starter/APU electrical equipment shall be subjected to a dielectric strength test which meets the requirements of MIL-E-25499.

4.5.21 Acceptance Tests. An acceptance test shall be run on each starter prior to delivery and shall consist of the following tests as a minimum.

4.5.21.1 Acceptance Data. An acceptance test data summary sheet shall be prepared for each starter unit. The following items shall be recorded as minimum:

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a. Date of test
b. Unit model designation
c. Unit serial number
d. Size of flywheel used
e. Ambient temperature (°F)
f. Ambient pressure (IN Hg Abs.)
g. Oil Temperature (°F)
h. Actual weight of unit, Dry (lbs)
i. Maximum unit vibration
j. Starter exhaust gas temperature, Maximum (°F)
k. Starter output torque at 1000 rpm and 2000 rpm output shaft speed
l. Cutoff speed (rpm)
m. APU performance at 3KVA and 6KVA

4.5.21.2 Examination of Product. The starter/APU shall be subjected to inspection to insure that all dimensions are within specification and drawing limits and that all parts are correctly assembled.

4.5.21.3 High Speed Components. Prior to conducting any operational starter tests, the following shall be accomplished on all high speed components:

Subject each part to x-ray inspection in accordance with MIL-STD-453. Any cracks or occulsions that will adversely affect starter performance or life shall be cause for rejection.

b. Subject each part to flourescent penetrant inspection in accordance with MIL-I-6866. Any cracks or occulsions that will adversely affect starter performance or life shall be cause for rejection.

c. Proof spin each turbine wheel for a minimum of 15 seconds at room temperature at the proof speed.

d. Upon completion of a, b, and c above, subject each turbine wheel to x-ray inspection to determine the extent of deformity, structural damage, growth of cracks or occlusions, or any new cracks shall be cause for rejection.

4.5.21.4 Leakage. Any leakage of liquids or gases from the starter, other than normal drainage, during acceptance cycling or overrunning shall be cause for rejection.

4.5.21.5 Performance. The starter/APU shall be subjected to one 40 second crank, two regular start cycles, one hour APU operation, and one translation cycle, as described in paragraph 4.5.2. The performance shall meet or exceed the requirements of

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figures 13 and 14 for starter performance and shall produce 6KW power output for APU performance.

4.5.21.6 Dielectric Strengh. The starter/APU electrical system shall be subjected to a test procedure to verify conformance with paragraph 3.6.24.4.

4.5.21.7 Burn-in. Each component which includes any electronic part, shall be subjected to this test. Temperature sensors shall be installed in the first component of each manufacturer's part number in the locations which are the last to stabilize in temperature as a result of following a temperature change, and subjected to a cycle of temperature as described below, starting with step b. The times required for steps c and a shall be recorded. Each component shall then be tested as follows: The component shall be placed in the temperature chamber, operated in normal (rated) conditions, and subjected to vibration in accordance with 4.5.6 procedure throughout the test. The component shall be subjected to cycles of temperature as follows until it has achieved three consecutive failure-free cycles:

a. The temperature of the chamber shall be reduced to and maintained at -67 +4°F until the temperature at all locations in the component are below -49°F (the capacity of the chamber shall be such that this shall not require more than one hour).

b. The temperature of the chamber shall be reduced to and maintained at -67°F for four hours,

c. The temperature of the chamber shall be increased to and maintained at 158 +4°F until the temperature at all locations in the component are above 140°F (the capacity of the chamber shall be such that this shall not require more than one hour).

d. The temperature of the chamber shall be maintained at 158 +4°F for a minimum of four hours,

The time for steps a and c shall be the same in testing subsequent components as that which was found necessary for the first component. Whenever there is any change in the manufacturer's part number for this component, or in the temperature chamber, temperature sensors shall be installed in the first component as described above and it shall be tested as described above to determine the time needed for steps a and c.

5. PACKAGING

5.1 Application. The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

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5.2 Preservation. All ports shall be sealed by closures conforming to MIL-C-5501. All exterior surfaces of the component shall be protected by corrosion-preventative compounds conforming to MIL-C-11796 or MIL-C-16173. The component shall then be wrapped or bagged in Grade A greaseproof paper conforming to MIL-B-121 and sealed with tape conforming to PPP-T-60.

5.3 Packaging.

5.3.1 Level A. Unless otherwise specified, each starter/APU shall be packaged for shipment in accordance with MIL-P-116. Method IIb of MIL-P-116 shall be employed for all starter/APUs designated as installation starter/APUs. Method IId of MIL-P-116 shall be employed for all starter/APUs designated as spare starter/APUs.

5.3.2 Level B. Starter/APU shall be packaged in accordance with standard commercial practice.

5.4 Packing. Shipping containers insofar as practicable, shall contain an identical number of articles, shall be of a uniform size, and shall be designed to enclose the contents in a snug, tightfitting manner. When packed for shipment, the gross weight of the exterior container shall not exceed 200 pounds. The above applies to 5.3.1 and 5.3.2.

5.4.1 Level A. Starter/APUs packaged in accordance with 5.3.1 shall be packed in an exterior shipping container in accordance with PPP-B-601 (overseas type) or PPP-B-621 (overseas type).

5.4.2 Level B. Starter/APUs packaged as specified in 5.3.1 shall be in an exterior shipping container conforming to PPP-B-601 (domestic type) or PPP-B-585.

5.4.3 Level C. The interior packages shall be placed in substantial commercial exterior shipping containers constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Except as specified herein, the container shall conform to the requirements of the Uniform Freight Classification Rules in effect at the time of shipment.

5.5 Marking. Each package and exterior shipping container shall be marked to comply with MIL-STD-129 and to include the following:

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Starter/APU, Aircraft Engine Gas
Turbine Type
Specification: MIL-S-85419 (AS)
Manufacturer's part number _____
Manufacturer's serial number _____
Government order to contract number _____
National Stock Number _____

NOTE : Complete the part number

6. NOTES

6.1 Intended Use. The starter/APU covered by this specification are intended for use in starting the F402-RR-404 and F402-RR-405 engines in the AV-8B aircraft.

6.2 Ordering Data.

6.2.1 Procurement Requirements. Contracts or orders should specify the following:

a. Title, number, and date of this specification.

b. Part number.

The levels of preservation, packaging, and packing (see Section 5).

6.3 Model Specification Preparation. A complete model specification conforming to the instructions for preparation contained herein shall be prepared and submitted by the supplier for approval by NAVAIR (AIR-536).

6.3.1 Instructions for Preparation. The model specification shall be prepared as follows:

a. The headings and numbering of sections and paragraphs shall correspond to those of this specification.

b. Paragraphs herein which are applicable as written shall be copied into the specification. Any change, addition or deletion shall be identified by placing an identifying mark or symbol in the margin.

c. Paragraphs herein which are not applicable to the particular starter/APU design shall have the words "not applicable" entered following the appropriate paragraph number and headings.

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d. Paragraphs requiring modification to define a particular design shall be modified only to the extent necessary to describe the characteristics of that particular starter/APU.

e. New requirements or additions shall be added as additional subparagraphs or as new paragraphs in logical sequence and location.

f. Items such as tables, figures, drawings, diagrams and appendices, shall be presented in complete form in the specification. Complete statements calling out these items shall be included in the text of the specification.

g. All curves shall be presented on graph paper having an adequate number of subdivisions to permit necessary interpolations.

6.3.2 Approval Status. The first page of the model specification shall contain information indicating the approval status of the specification in the upper left hand corner-. If approval has been obtained, enter the word “approved” and cite the communication which conveys the approval. If the specification is not yet approved, enter the words “not approved”.

6.3.3 Specification Dates. The date of the specification shall be provided in the upper right hand corner.

6.3.4 Revisions. When revisions are made, they shall be designated by the use of a letter following the specification number with a revision date therefor, which shall be shown on page 1 only of the model specification. Only the specification number and revision suffix letter, if applicable, shall be shown on subsequent pages. Revision by preparation of revised pages for insertion into previously submitted copies of the model specification shall not be used.

Preparing Activity
Navy - AS

(PROJECT NO. 2995-N132)

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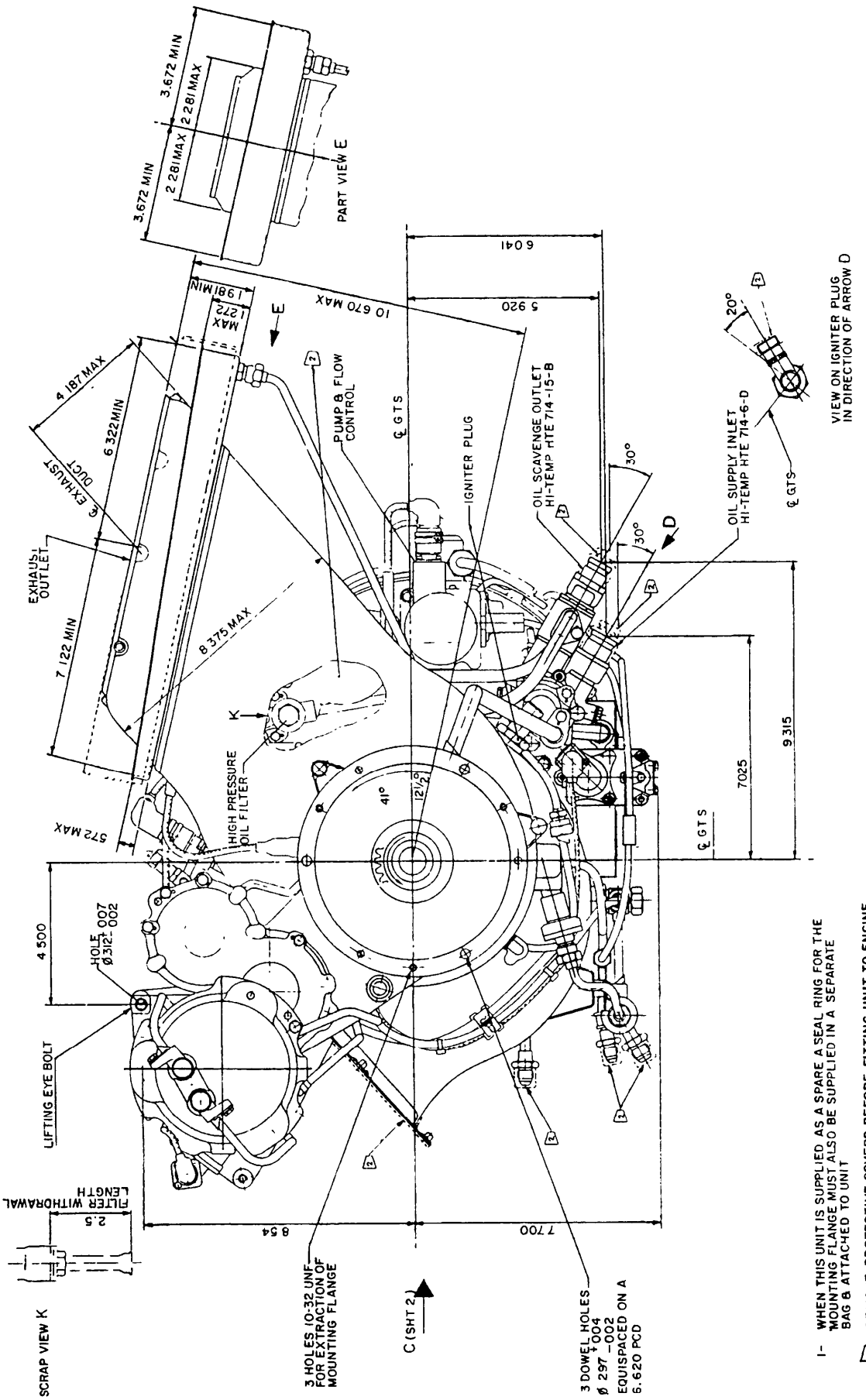


FIGURE 1A. Starter/APU outline drawing (drive end view).

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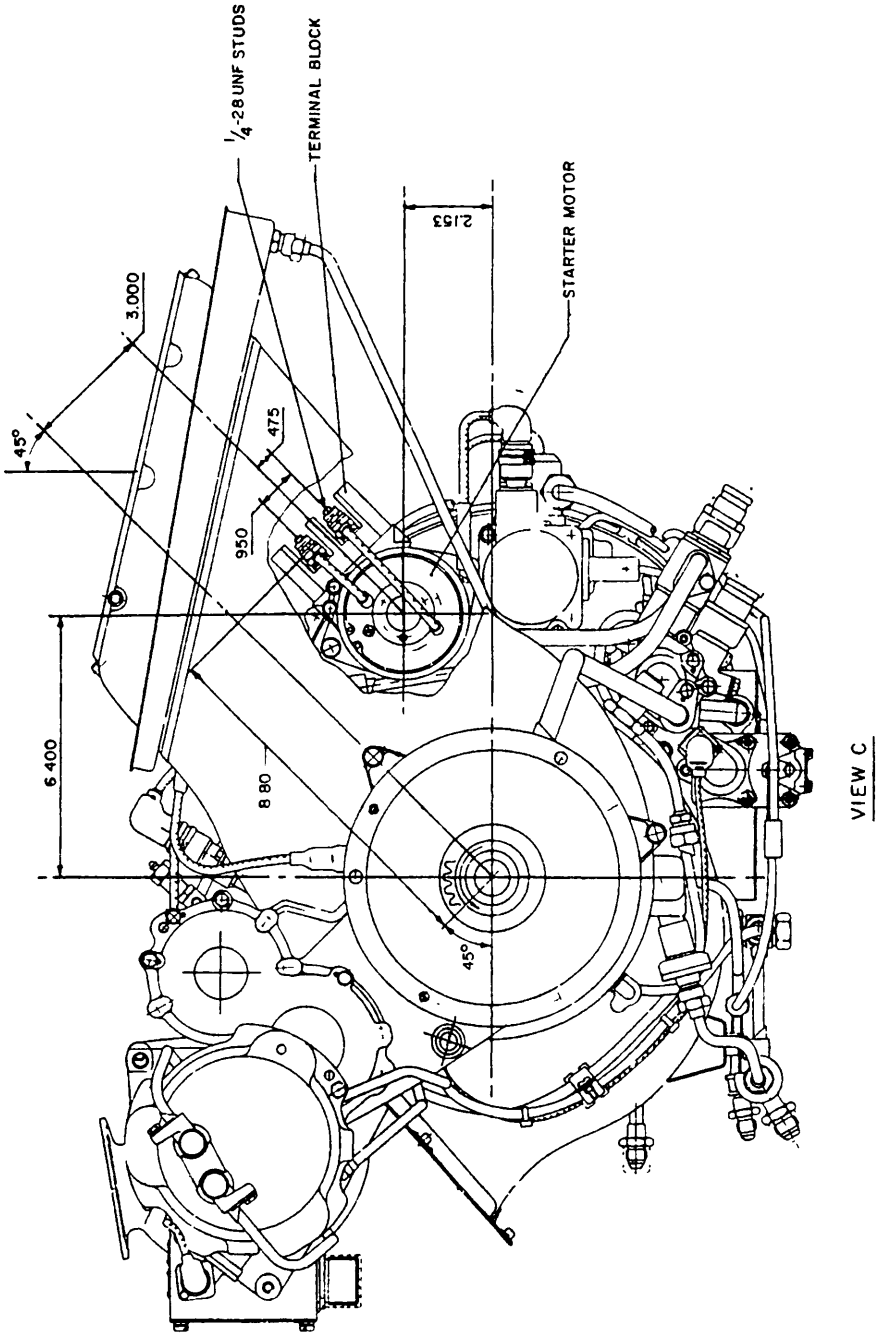


FIGURE 1B. Starter/APU outline drawing (drive end view with starter details).

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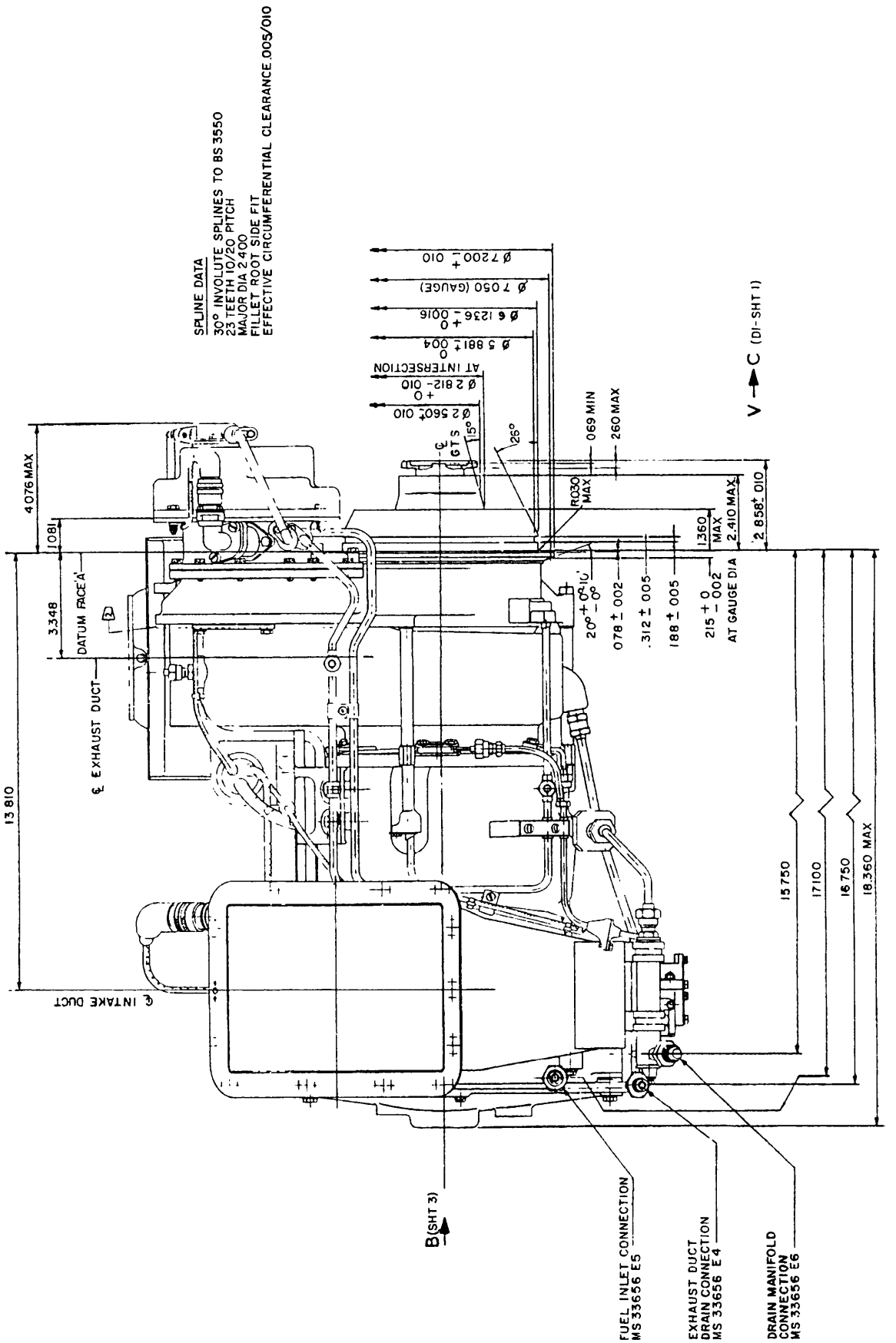
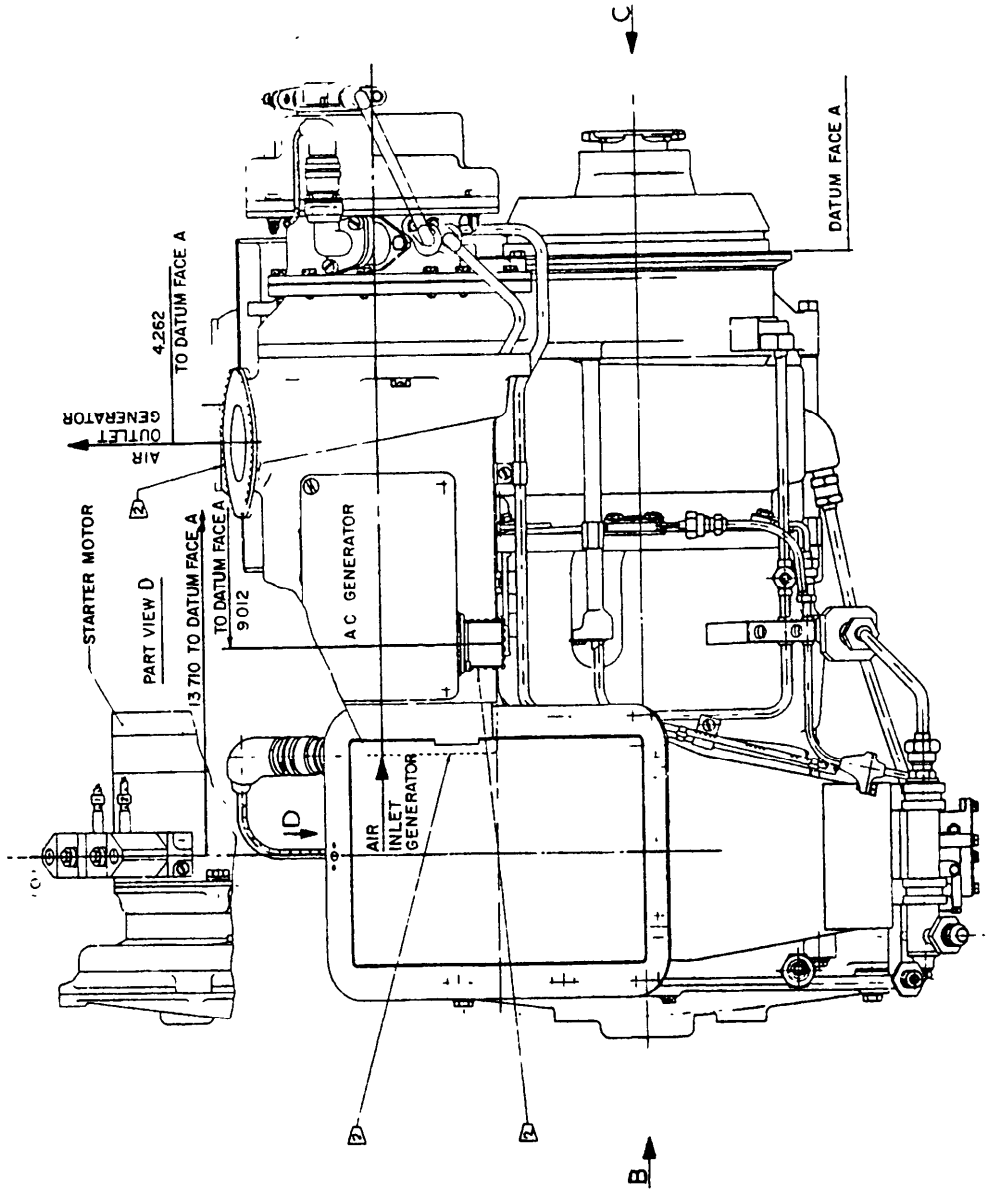


FIGURE 2A. Starter/APU outline drawing (top view).

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NOTES

1- WHEN THIS UNIT IS SUPPLIED AS A SPARE A SEAL RING FOR THE MOUNTING FLANGE MUST ALSO BE SUPPLIED IN A SEPARATE BAG AND ATTACHED TO UNIT.

[2] REMOVE PROTECTIVE COVERS BEFORE FITTING UNIT TO ENGINE.

FIGURE 2B. Starter/APU outline drawing (top view rotated).

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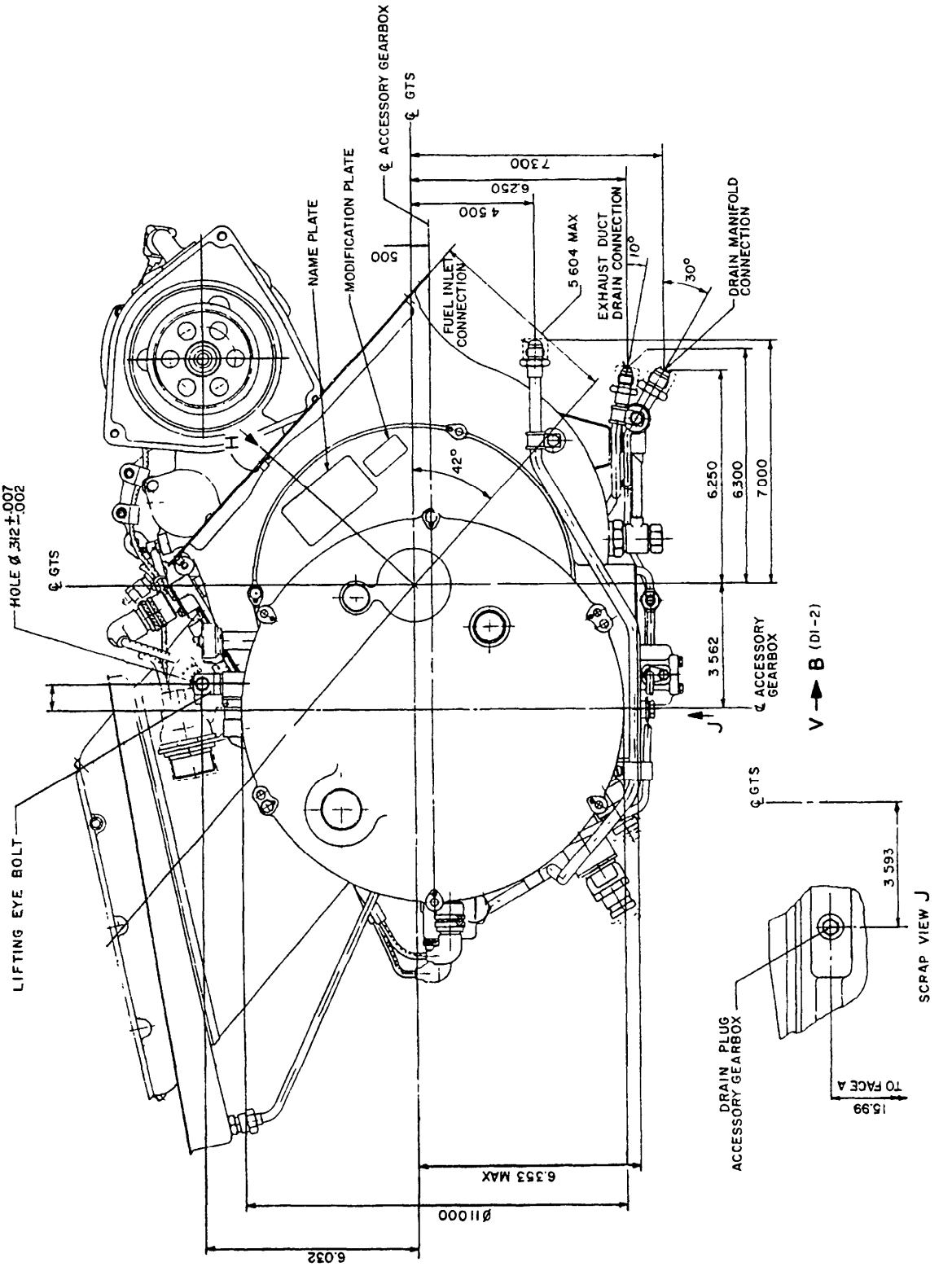


FIGURE 3A. Starter/APU outline drawing (anti-drive end view without generator).

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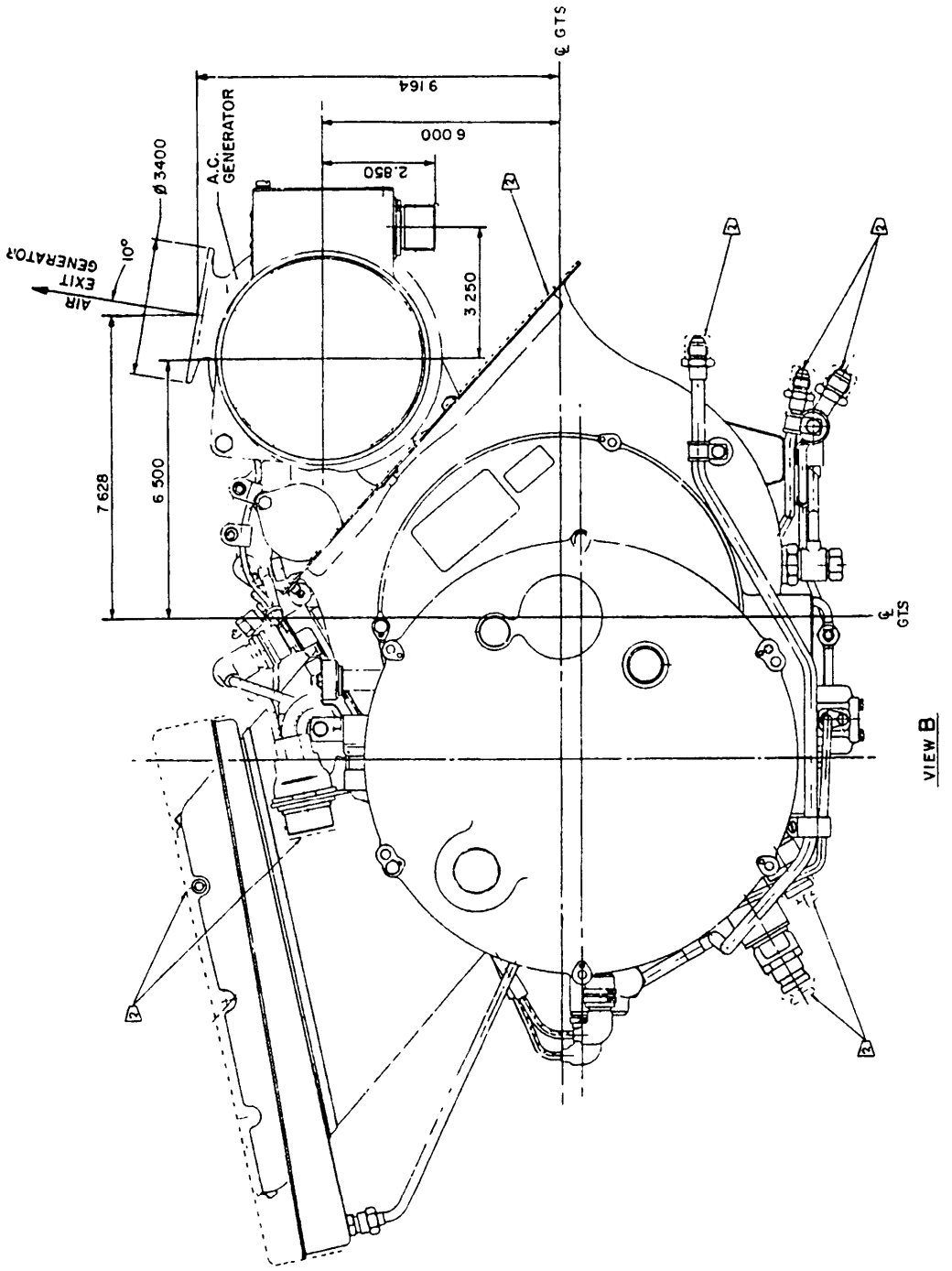


FIGURE 3B. Starter/APU outline drawing (anti-drive end view with generator).

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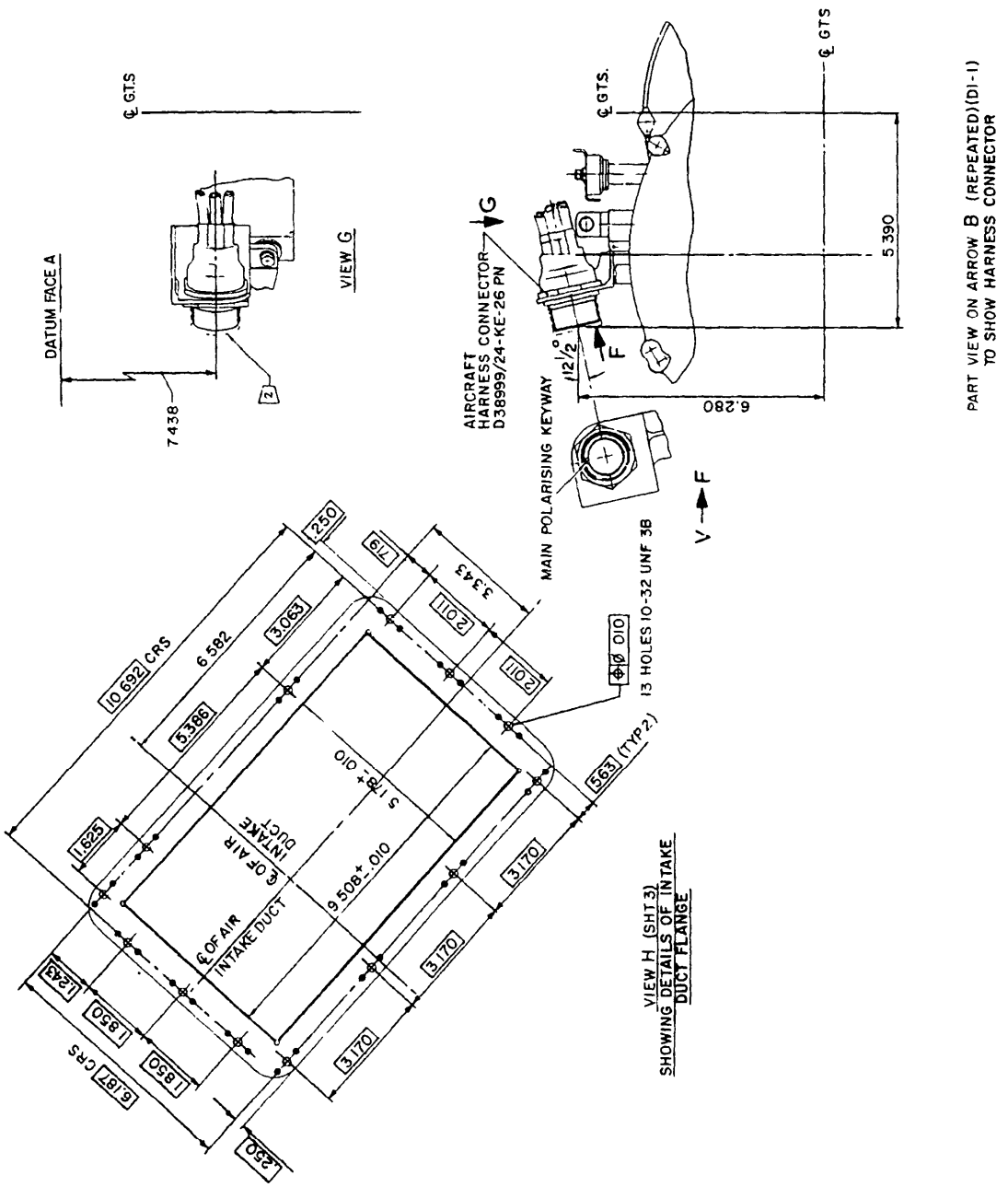


FIGURE 4. Starter/APU outline drawing (intake duct details).

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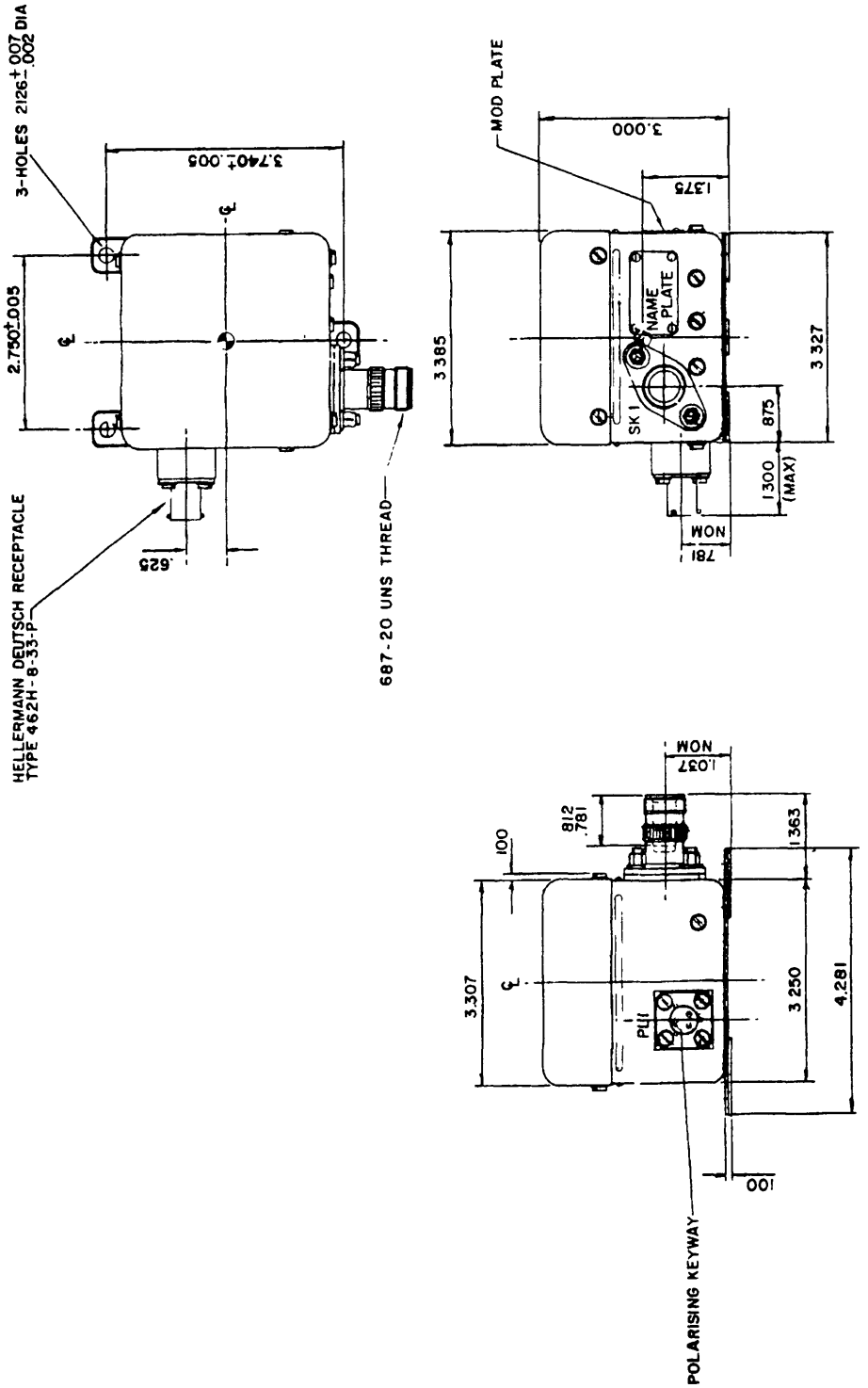


FIGURE 5. Ignition unit outline drawing.

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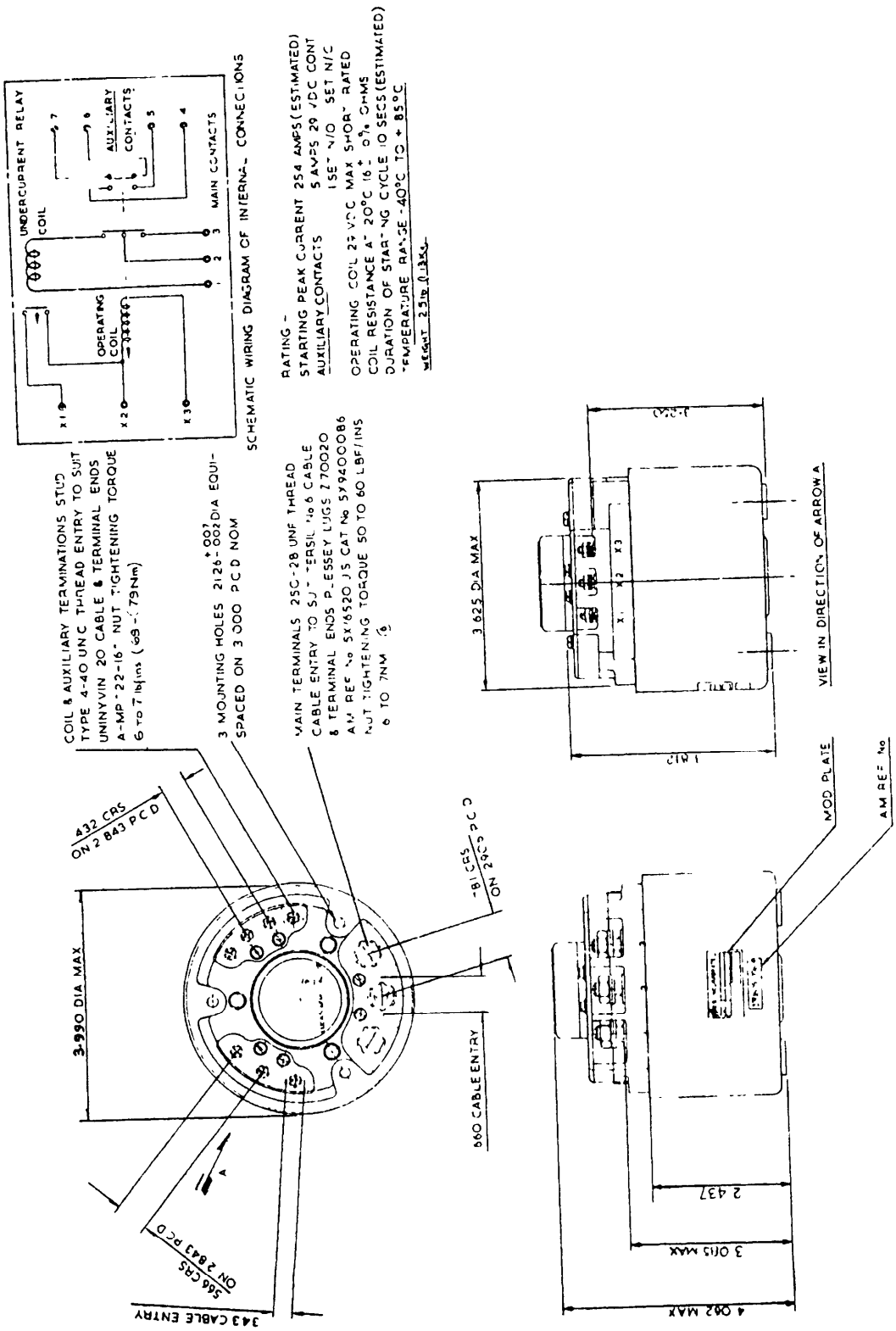


FIGURE 6. Starter contactor outline drawing.

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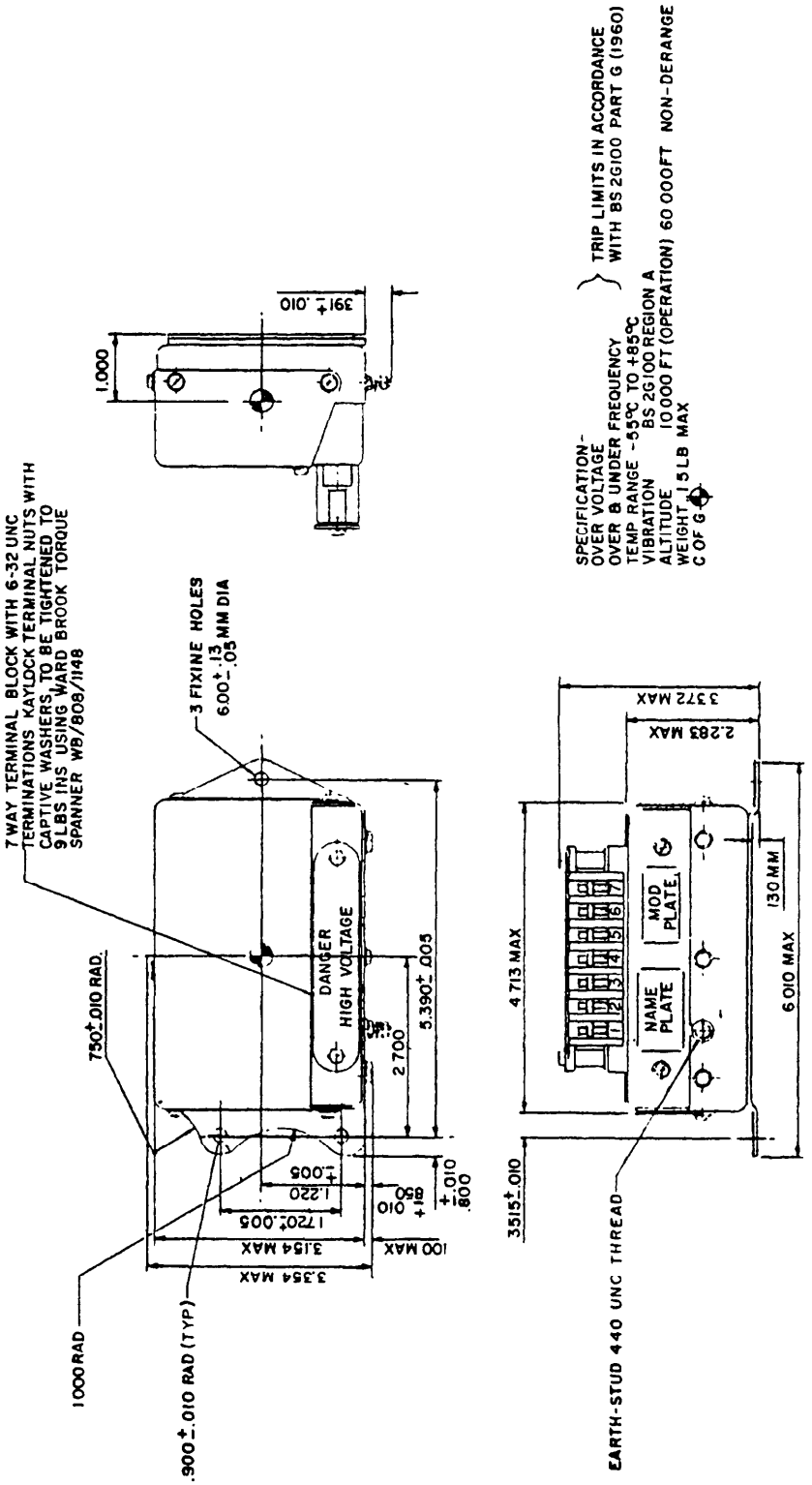


FIGURE 7. Generator control unit outline drawing.

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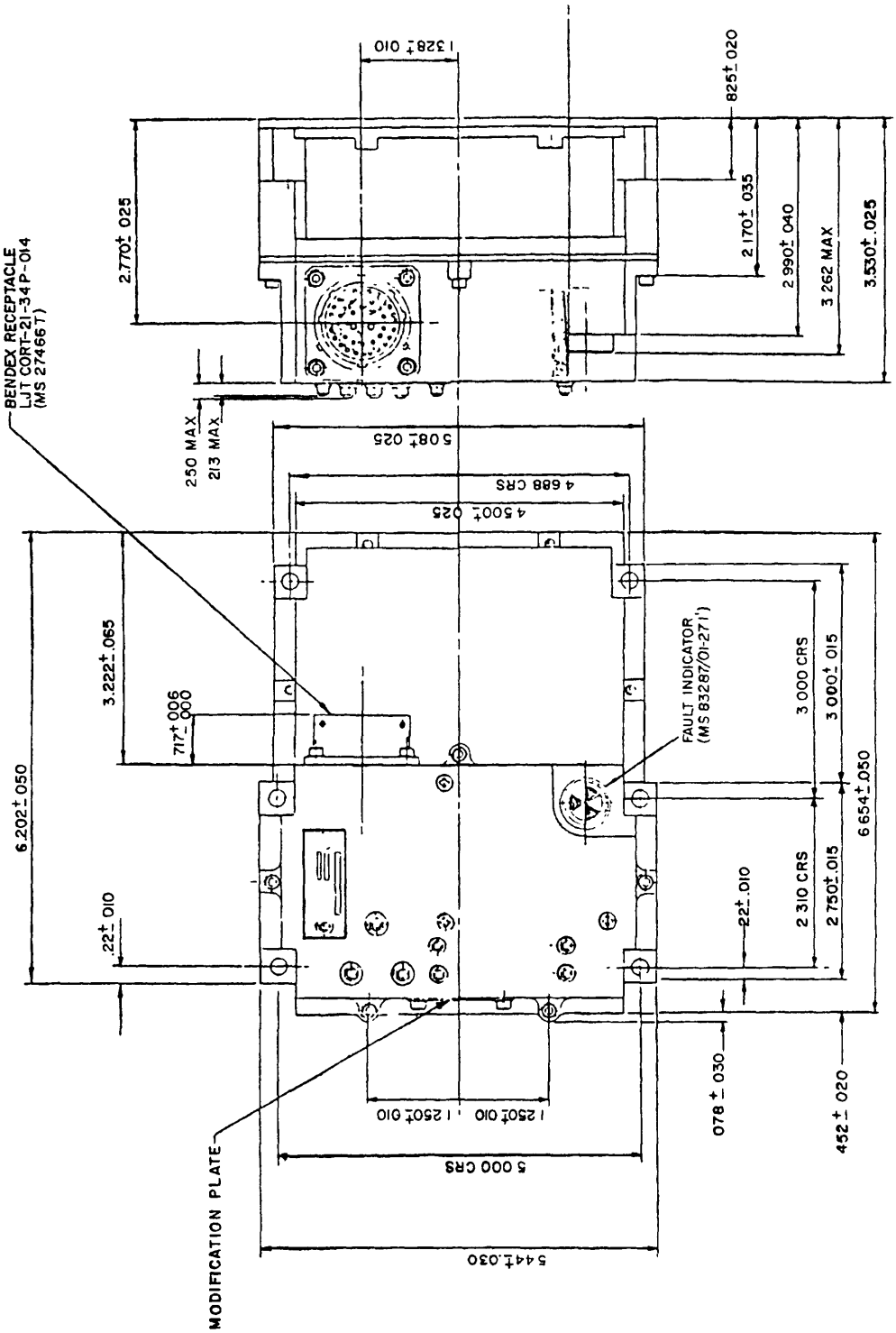


FIGURE 8. Sequence and safety control outline drawing.

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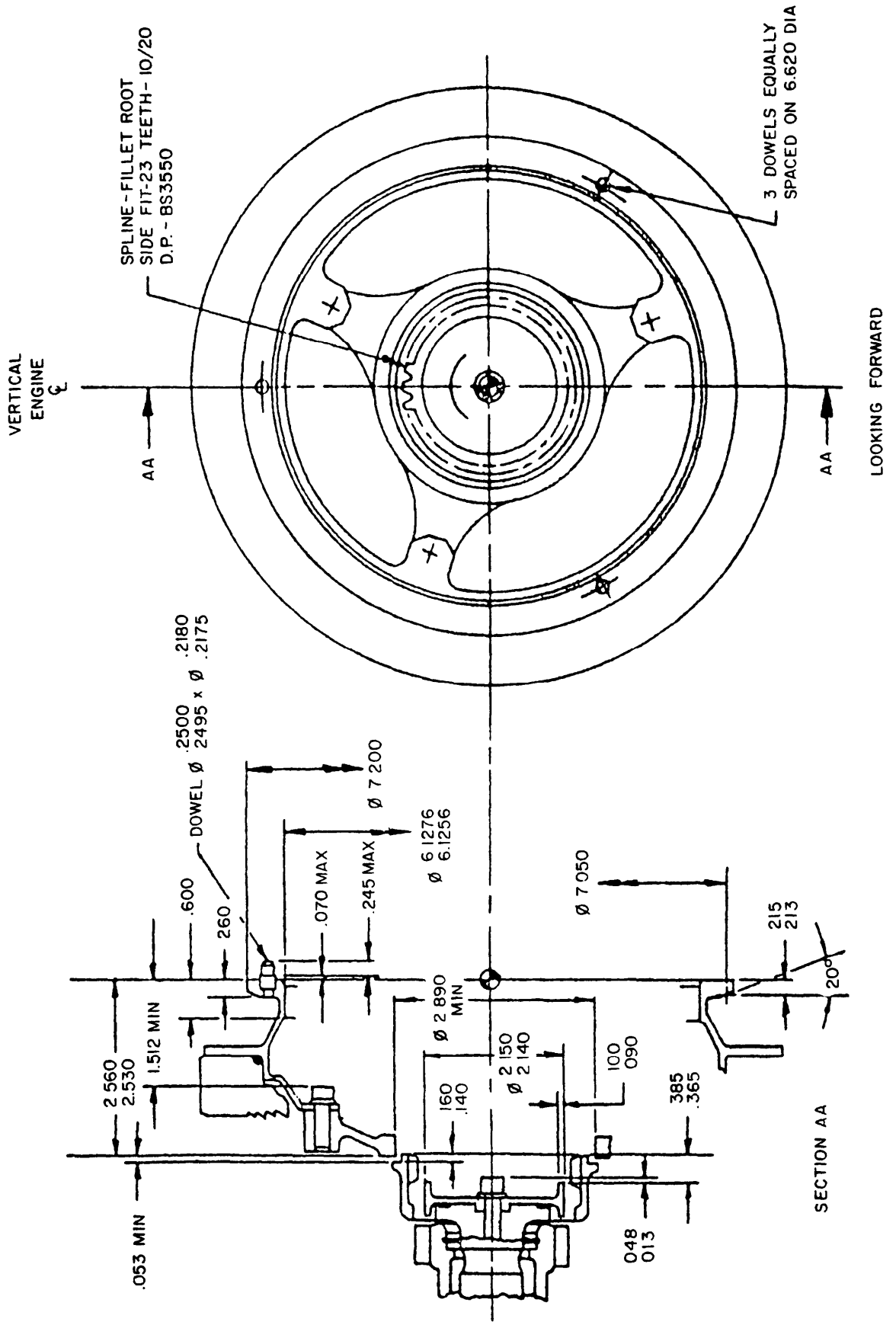


FIGURE 9. Starter drive interface.

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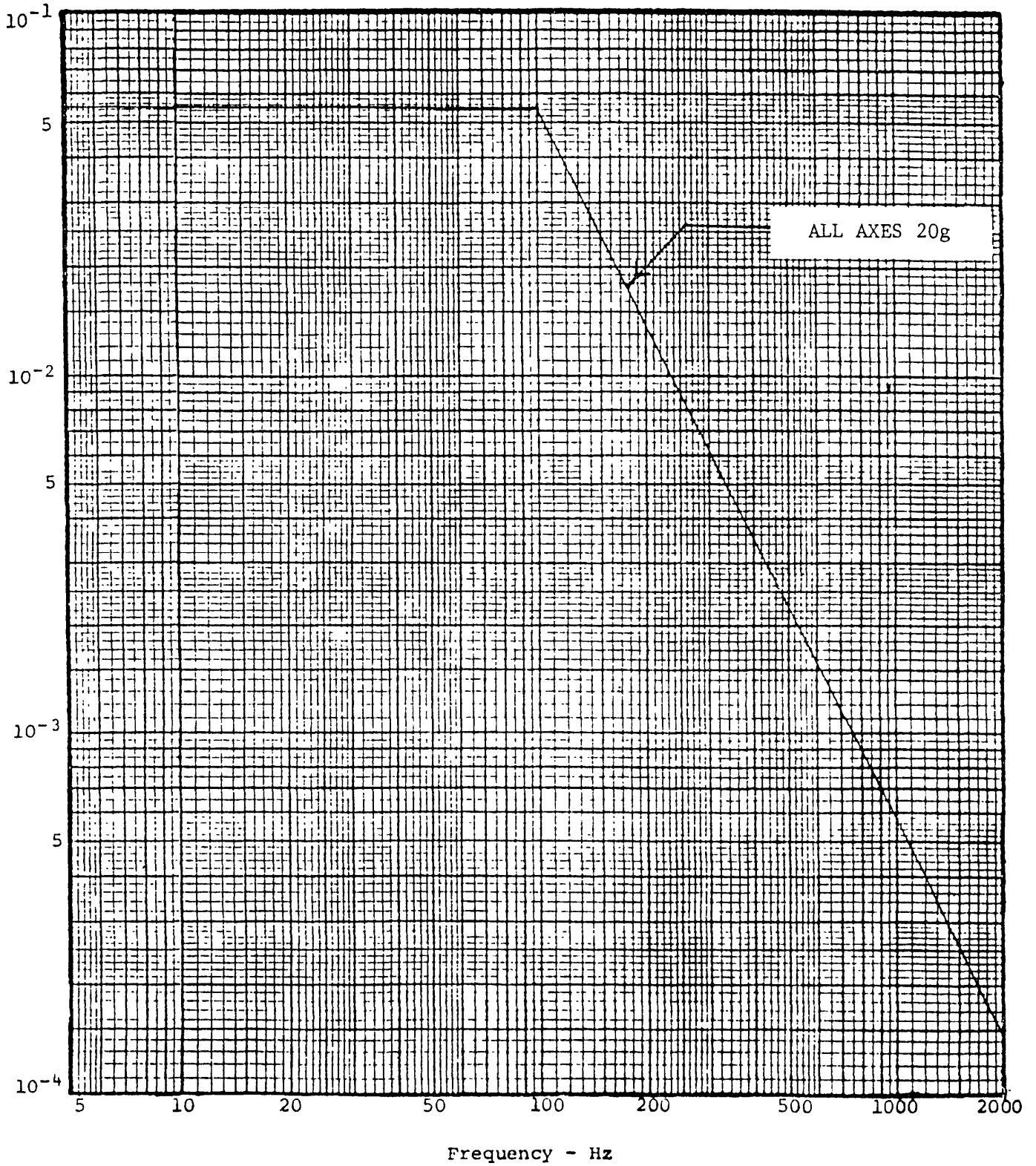


FIGURE 10. Starter vibration environment.

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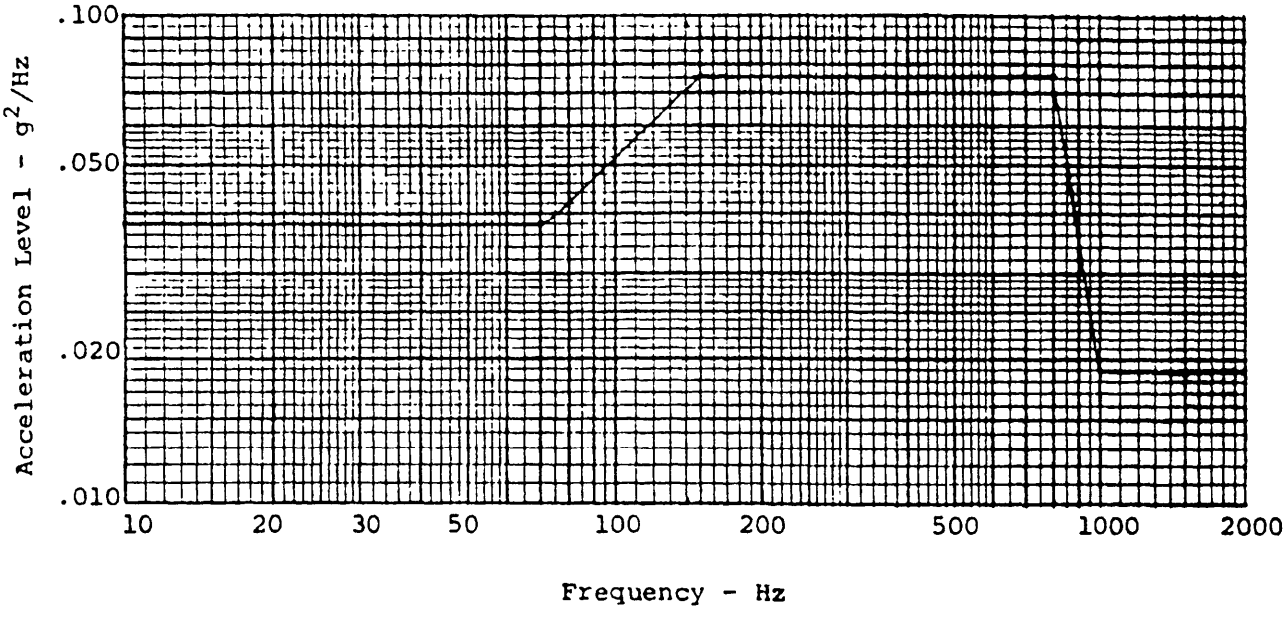


FIGURE 11. Vibration environment starter electrical controls.

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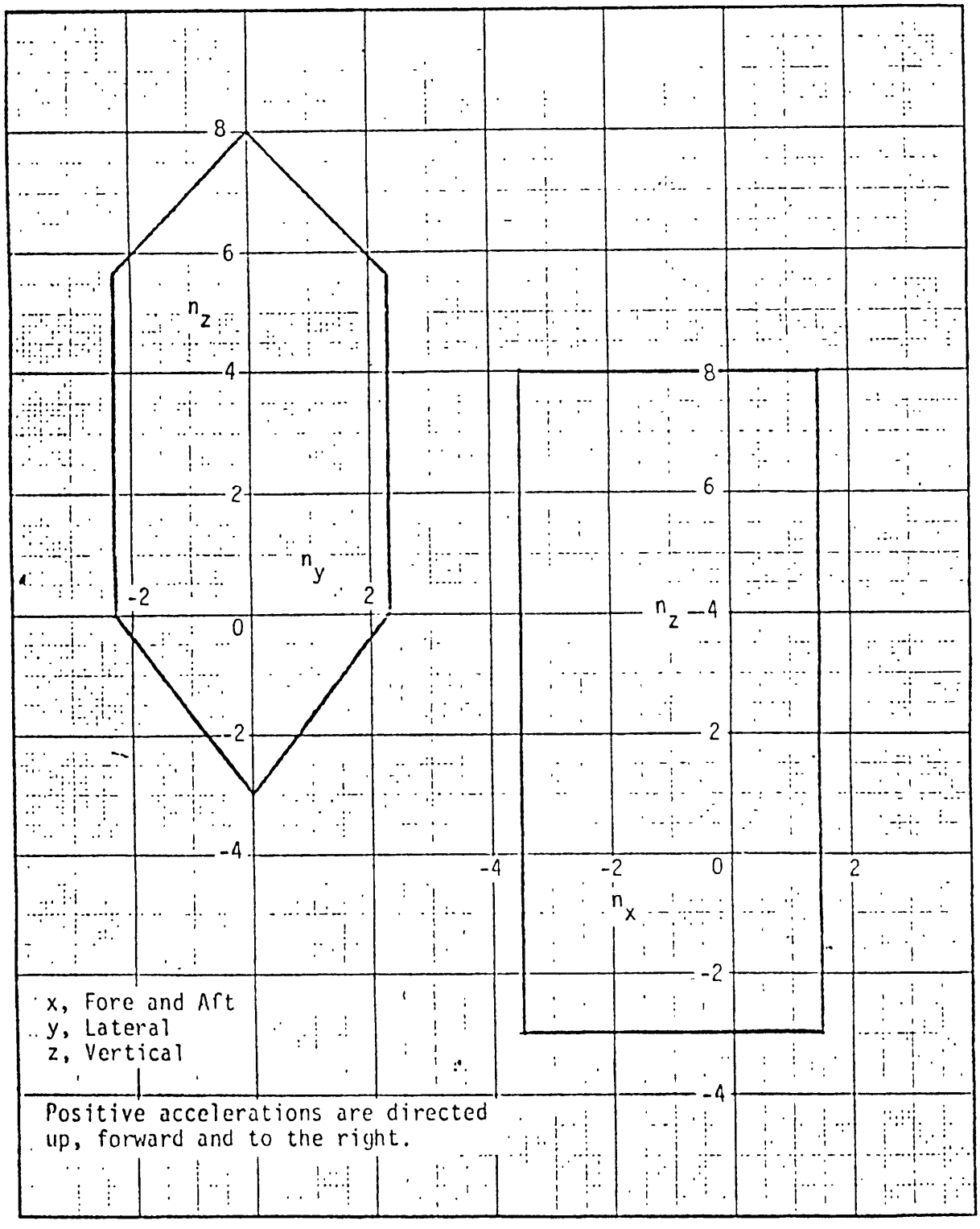


FIGURE 12. Starter design limit load factors.

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A -15°F (-26°C) SEA LEVEL

B +59°F (+15°C) SEA LEVEL

C +122°F (+50°C) SEA LEVEL

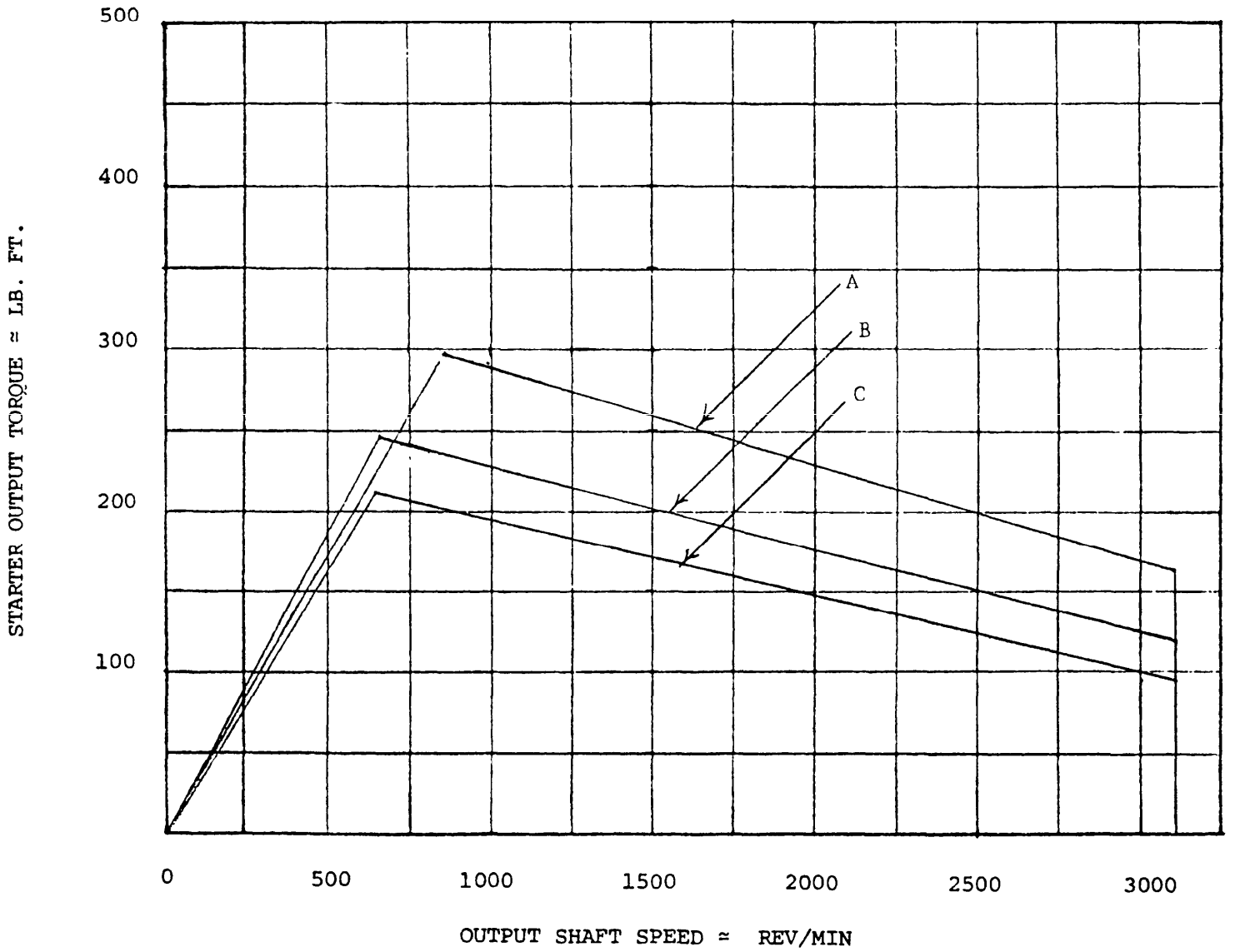


FIGURE 13. Starter torque vs. speed.

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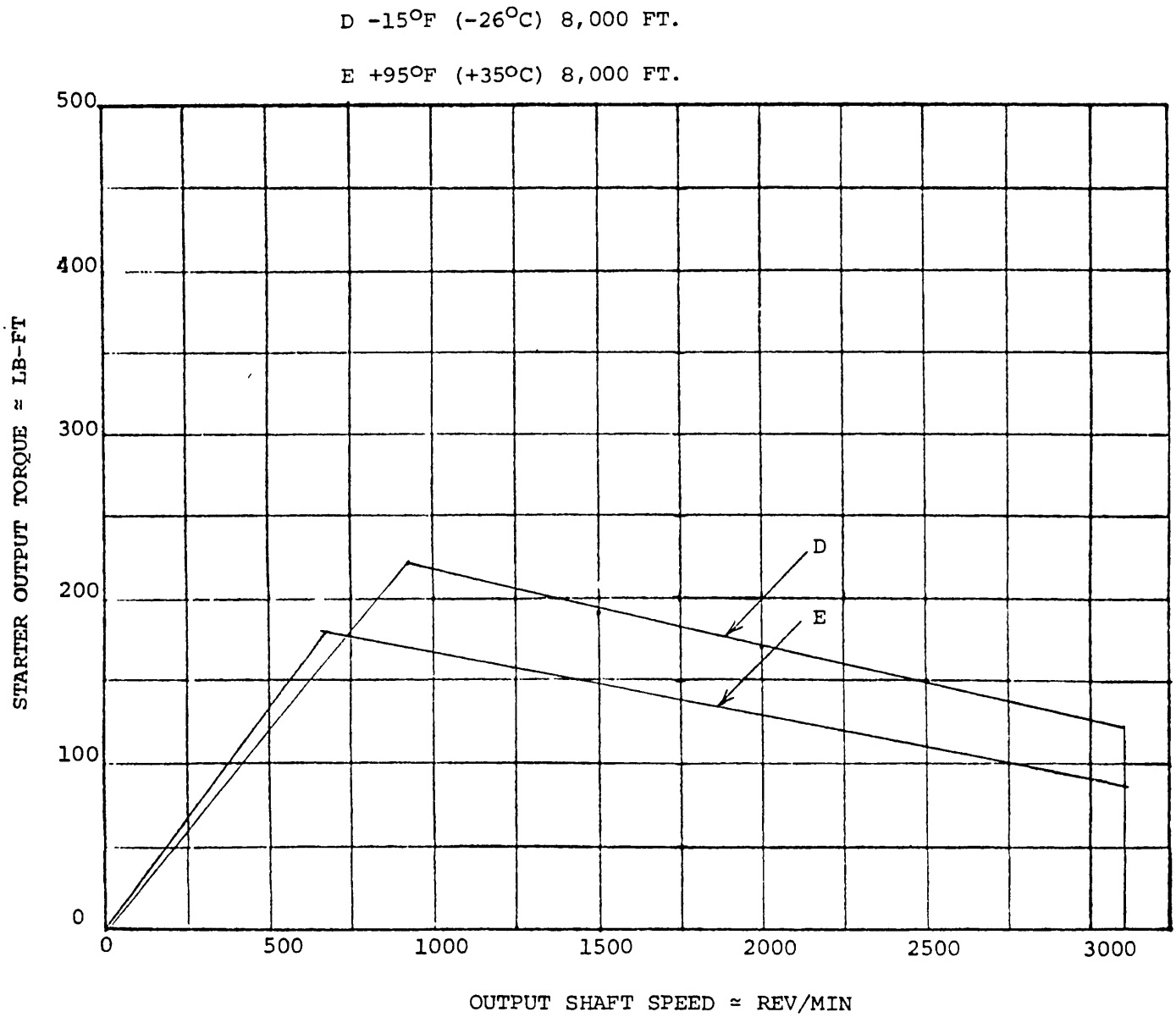


FIGURE 14. Starter torque vs. speed.

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